
بسم الله الرحمن الرحيم
In the name of Allah the most beneficent the most merciful

SPECTRAL REFLECTANCE

SPECTRAL REFLECTANCE

- **The average amount of incident radiation reflected by a surface at some wavelength interval is called '*Spectral reflectance* ρ_λ" and normally expressed as a percentage**

$$\rho_\lambda = E_R(\lambda) / E_I(\lambda)$$

$$= \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ incident upon the object}} \times 100$$

ALBEDO and SPECTRAL REFLECTANCE

- ***Albedo**, is defined as the ratio of the amount of electromagnetic energy reflected by a surface to the amount of energy incident upon it.*
- **Usually albedo is averaged over the visible range of the Electro-Magnetic spectrum**
- **Reflectance relates to a specific wavelength (or a specific band of a satellite)**

SPECTRAL REFLECTANCE CURVE /PROFILE

- Graph of spectral reflectance of an object as a function of wavelength is termed as “Spectral reflectance curve/profile”
- Curve gives us insight into the spectral characteristics of an object and has a strong influence on the choice of wavelengths in which remote sensing data is acquired for particular applications
- Reflectance curves are made for the optical part of the electromagnetic spectrum (up to 2.5-3 μm)

SPECTRAL REFLECTANCE

- To relate different kinds of surfaces and their spectral signatures (to how they look like in a satellite image), efforts are made to collect and store of typical curves in spectral libraries
- Reflectance measurements can be carried out in a laboratory or in the field using a Field Spectrometer



SPECTRAL RESPONSE

- A basic assumption of RS is that specific targets have individual manner of interacting with the incident radiations, and is described as " SPECTRAL RESPONSE" of a target
- Specific targets such as Soils of different types, water with varying degrees of impurities and depths, rocks of different lithologies and vegetation of various species have different spectral response
- Spectral response may very from time to time during year such as vegetation (crops) grows

REFLECTANCE

- A blue object appears blue because it
  - absorbs all the wavelengths except the one corresponding to blue color
- A purple object reflects both
  - red and blue
- A green roof can be distinguished from a red roof because incident light is
  - reflected highest in green portion of the spectrum from the green roof and highest from red portion of the spectrum from the red roof

SPECTRAL REFLECTANCE OF WATER

INTERACTION OF EMR WITH WATER

- Possible reflectance from Water bodies
  - Surface
  - Volume
  - Bottom

INTERACTION OF EMR WITH WATER

- Spectral reflectance of water is effected by
  - Presence and concentration of dissolved matter
  - Depth of the water body (reflection form the bottom)
  - The wavelength
  - Surface conditions (calm/smooth, rough)
  - Angle of observation and illumination

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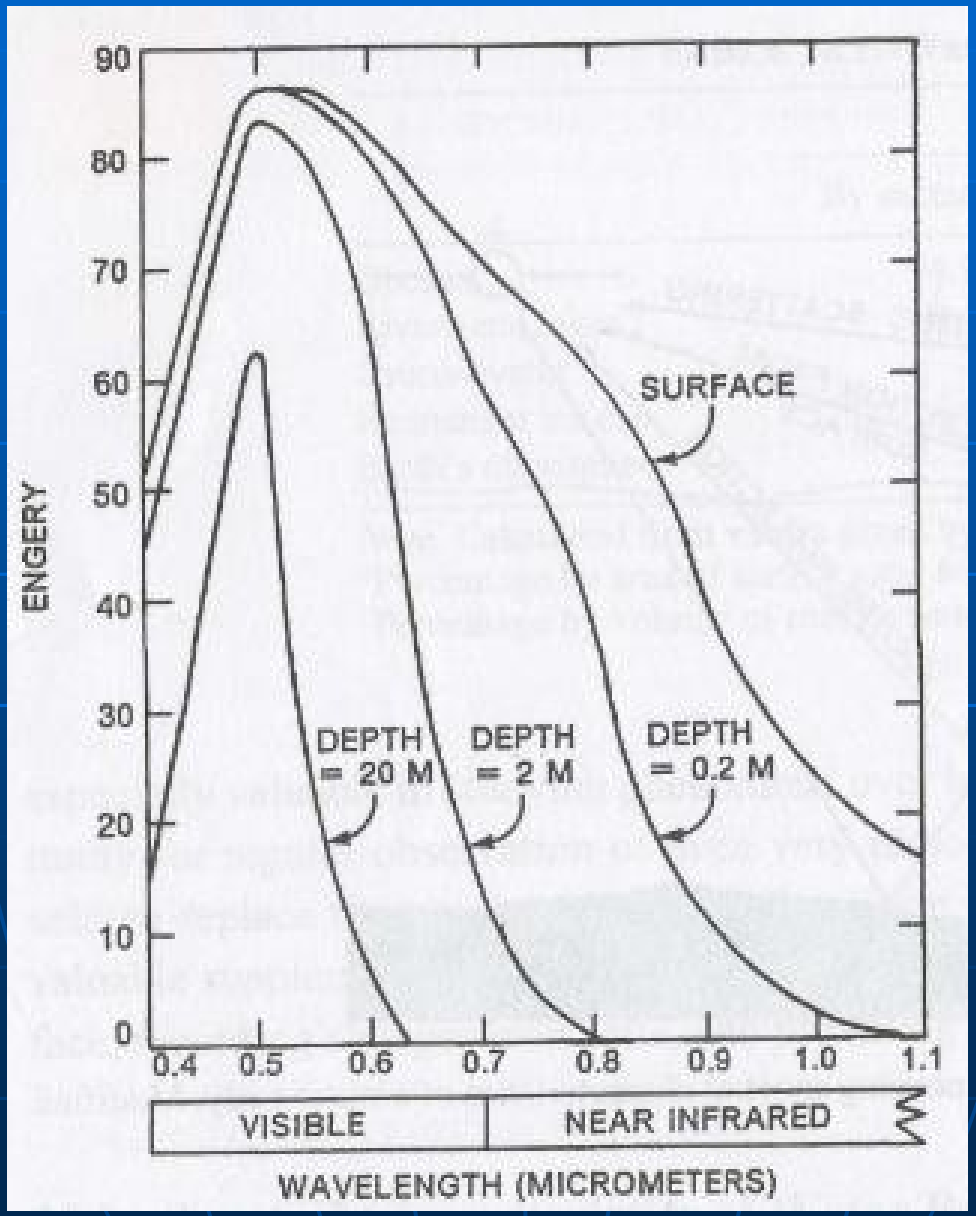
INTERACTION OF EMR WITH WATER

- Surface reflectance carry less info about water and more about the surface conditions, such as wind and waves effects
- Spectral reflectance of water body depends upon energy scattered or reflected with in the water (occurs at a range of depths)
- Deep water bodies with absence of impurities will appear blue/green (Scattering by particles that are small relative to the wavelength, cause shorter WL to scatter more)

INTRACTION OF EMR WITH WATER

- In clear water, it is possible to estimate the depth from the intensity of visible radiation particularly blue light reflected from the bottom
- At greater depth, all visible radiation is absorbed and water bodies appear dark
- In NIR region, water absorbs all the incident energy and is the only natural material with this property, so water bodies can be distinguished easily from other surface features in this part of EM spectrum

INTRACTION OF EMR WITH WATER



Moore (1978, p. 458).

INTRACTION OF EMR WITH WATER

- Sediment's concentration change spectral properties (Turbid waters)
- Brightness increases in Visible part, no more look as a dark object
- Increase in sediment contents shift the peak reflectance from shorter WL (blue) towards green and red, change of color
- Higher concentration, color may be same as of the sediments
- Degree of scattering is proportional to the concentration of particles

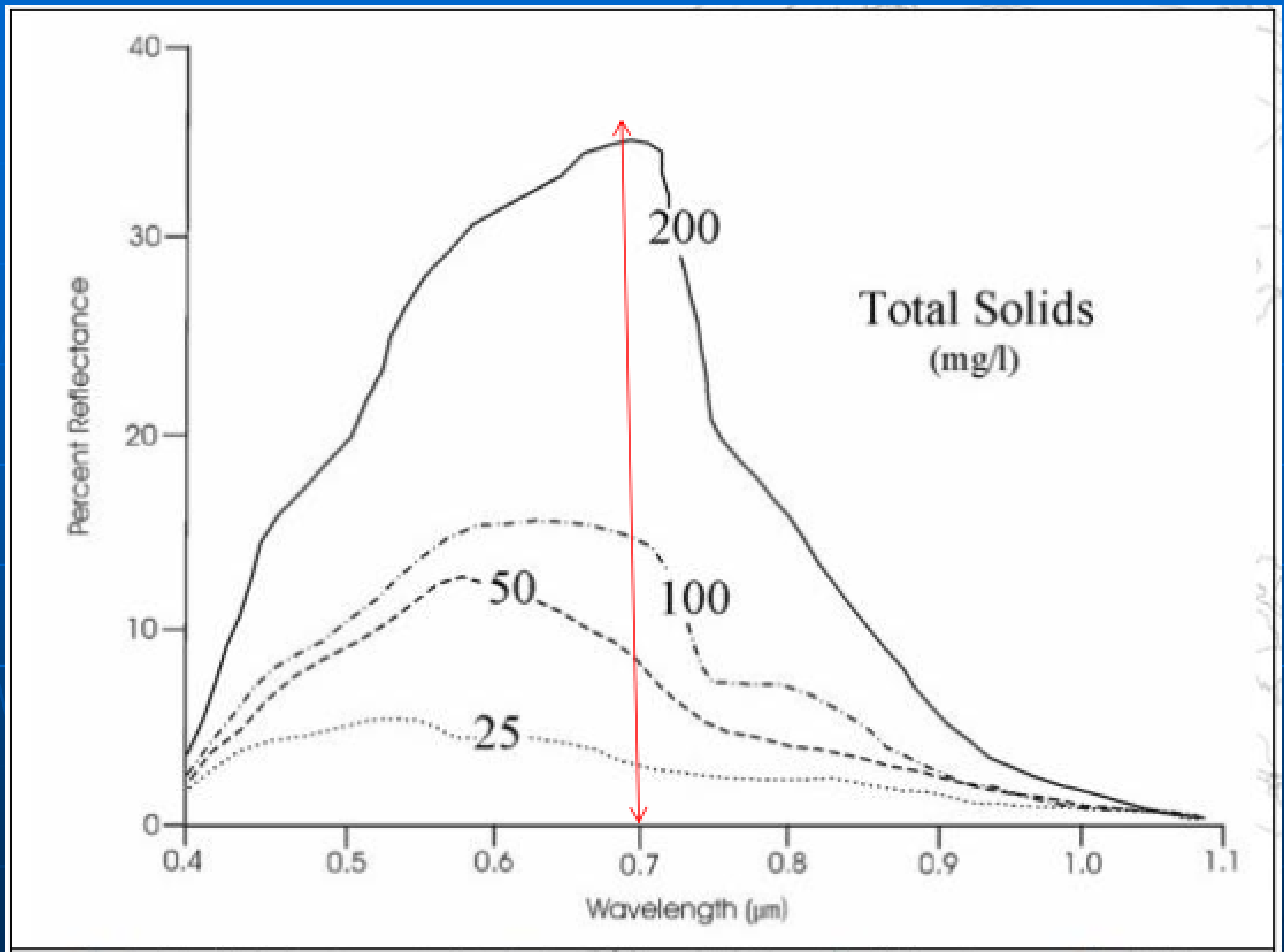
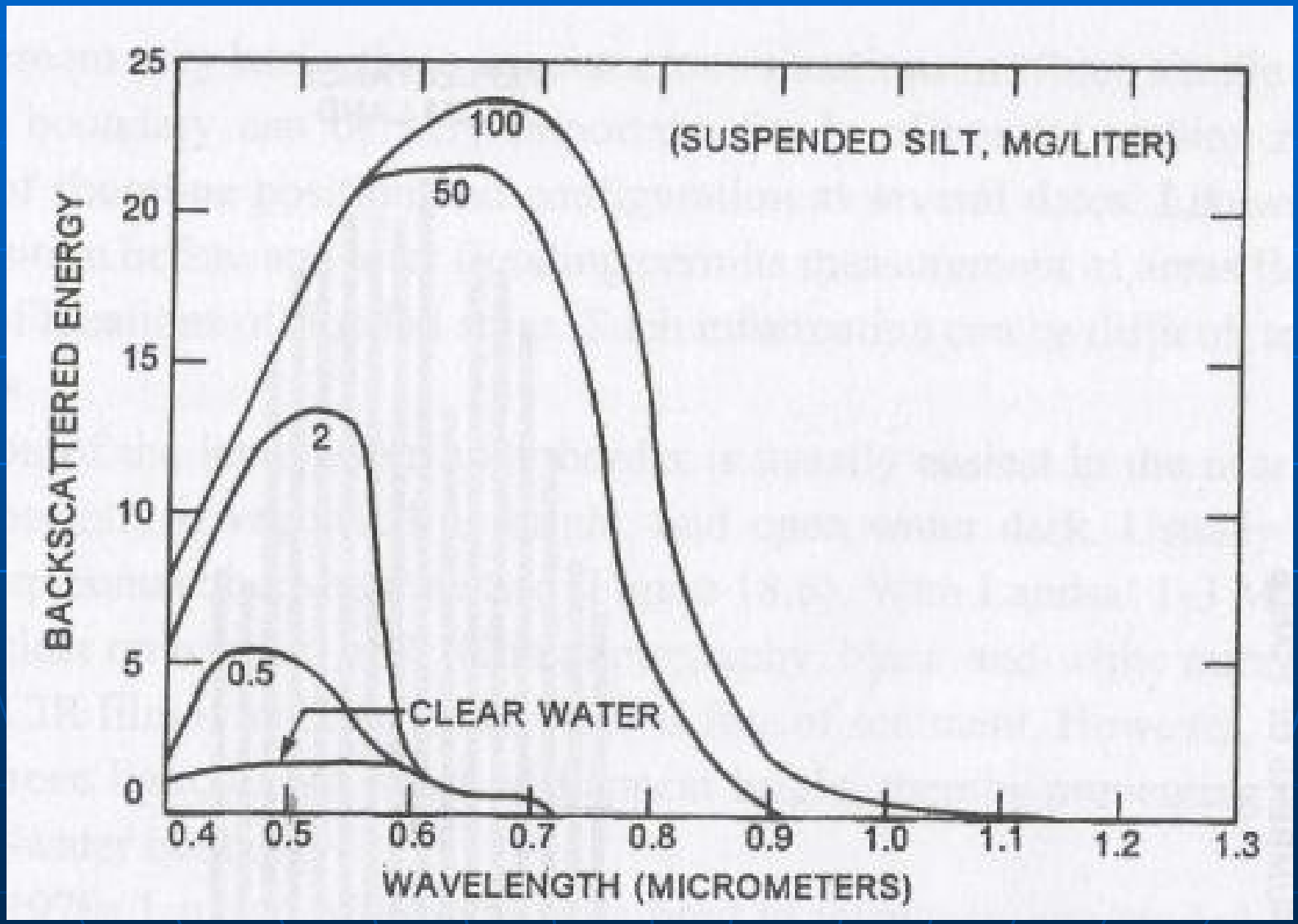
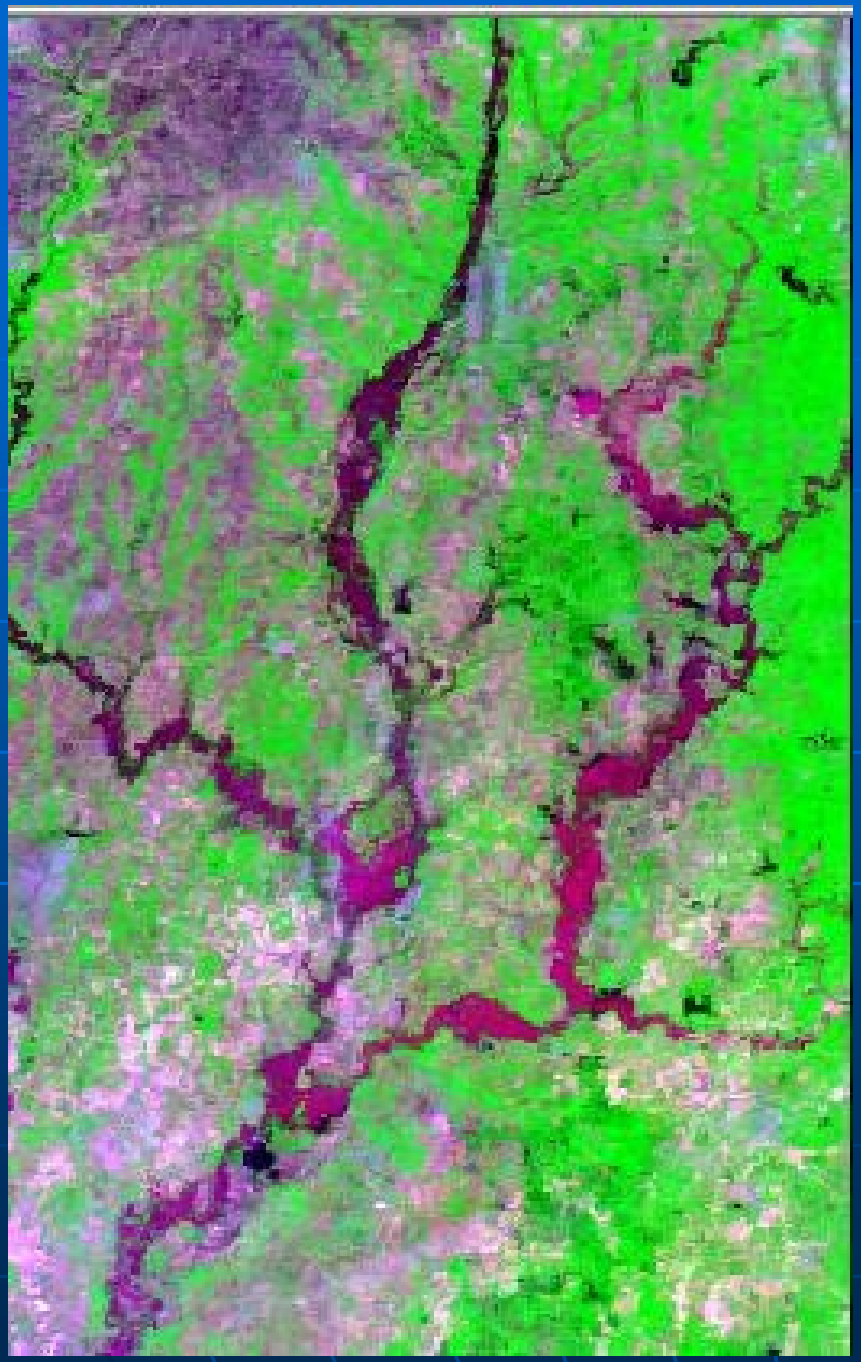


Fig 9: Variation in the spectral reflectance characteristics of turbid water according to the content of suspended solids.

INTRACTION OF EMR WITH WATER






Turbid water

Flood Water



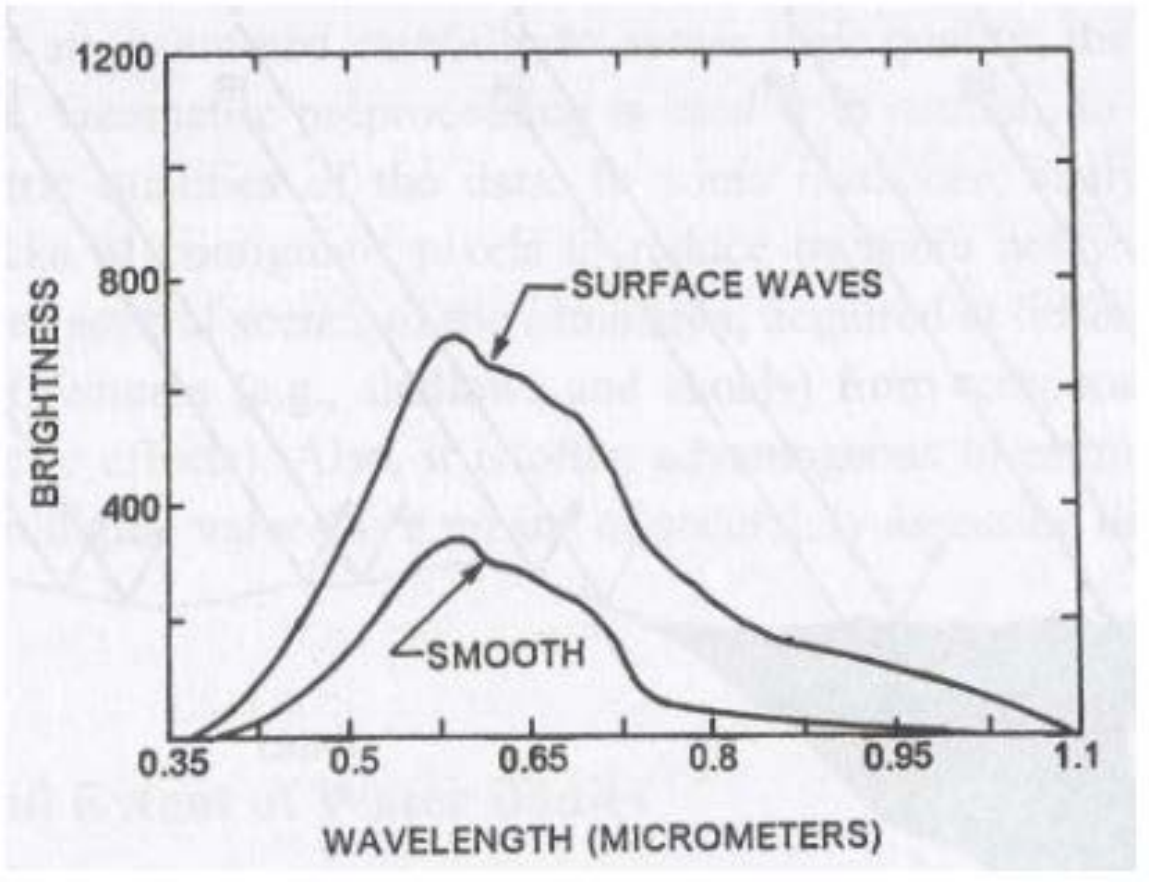
Turbid water



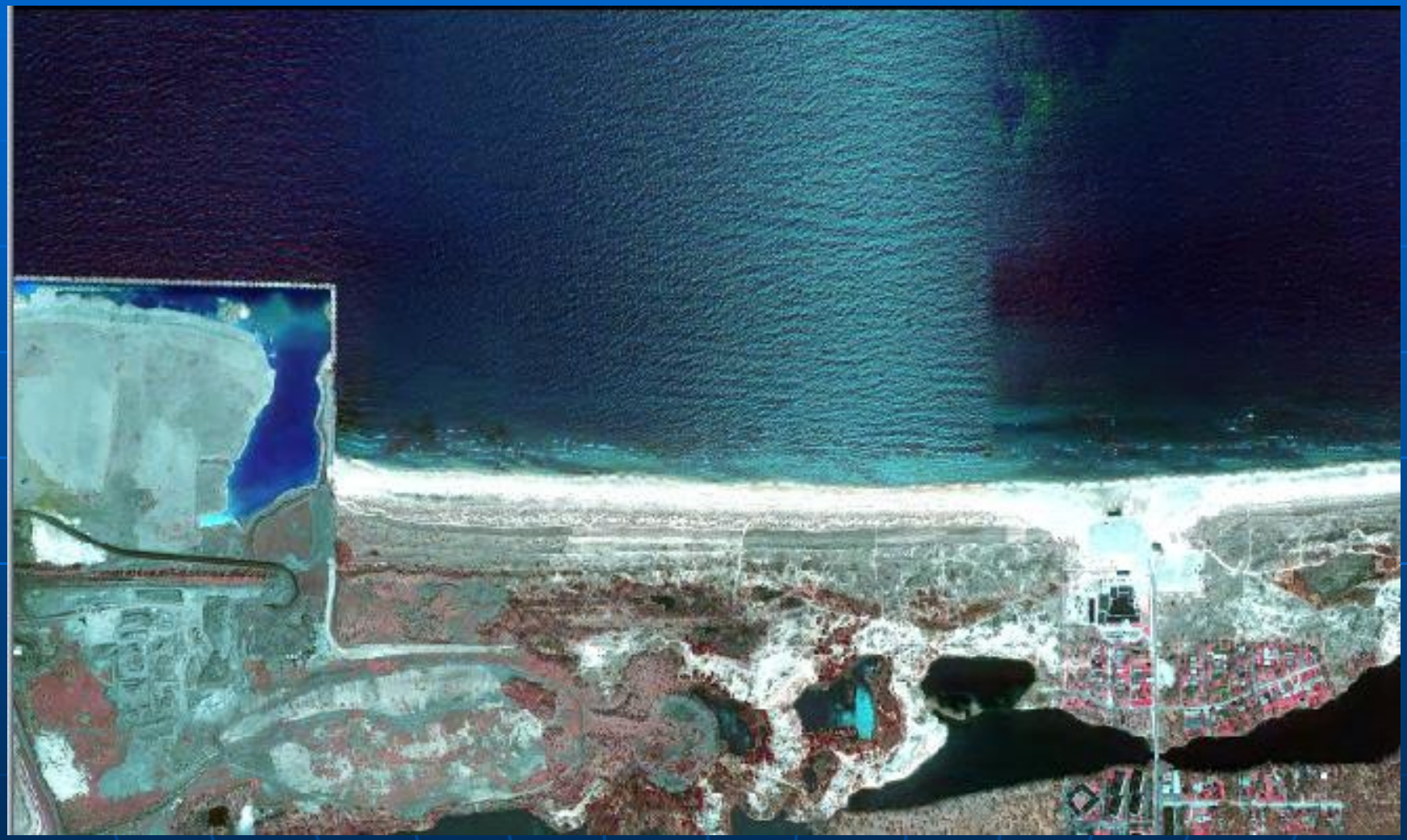


Roughness of water surface

- Roughness due to wave action increases the brightness (more in visible)
- Calm, smooth water surfaces direct only volume-reflected radiation to the sensor



Wind and Wave's effect on water surface reflectance



Wind and Wave's effect on water surface reflectance

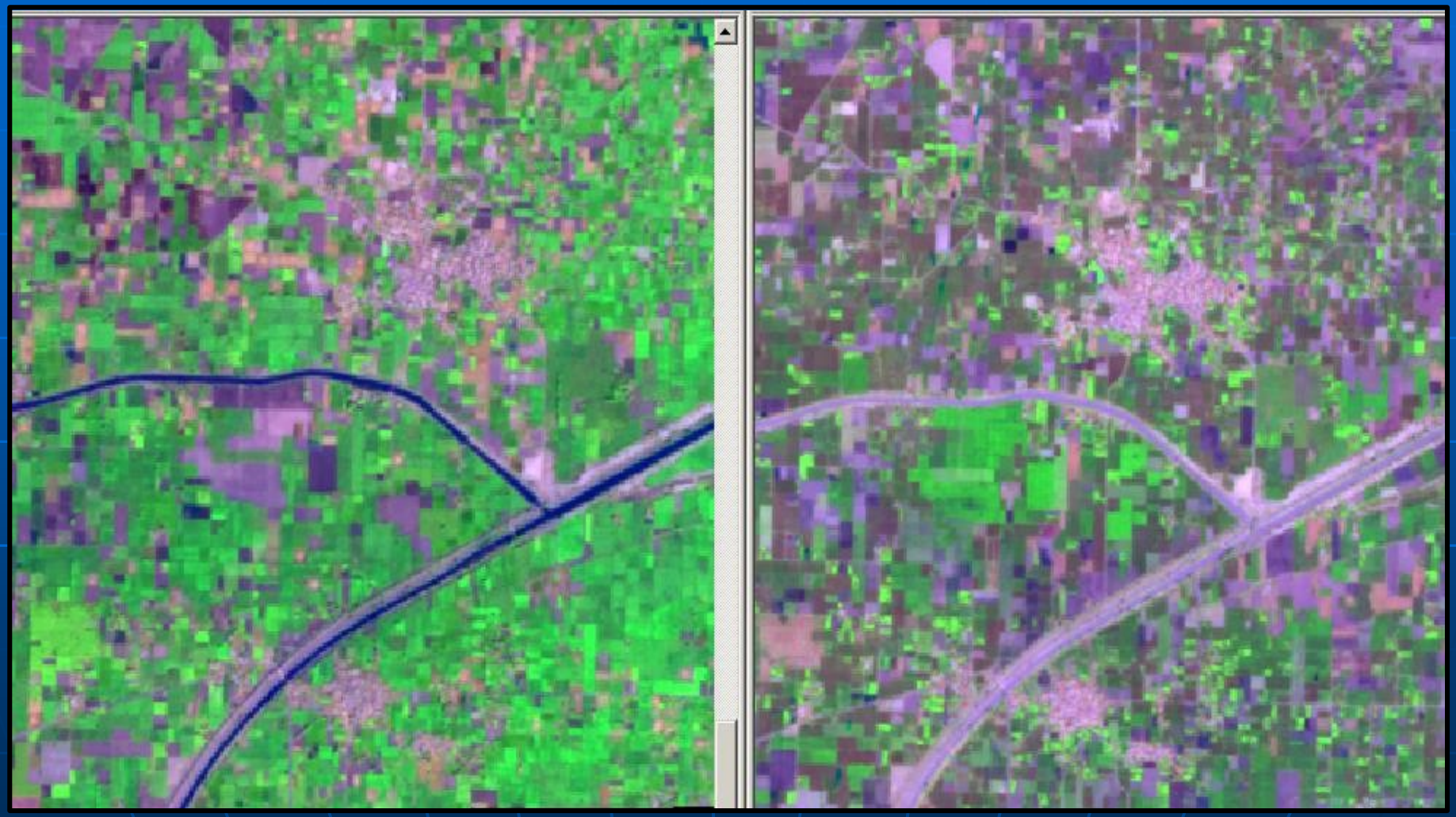


Spectral Reflectance –Temporal Effects



Apr 05

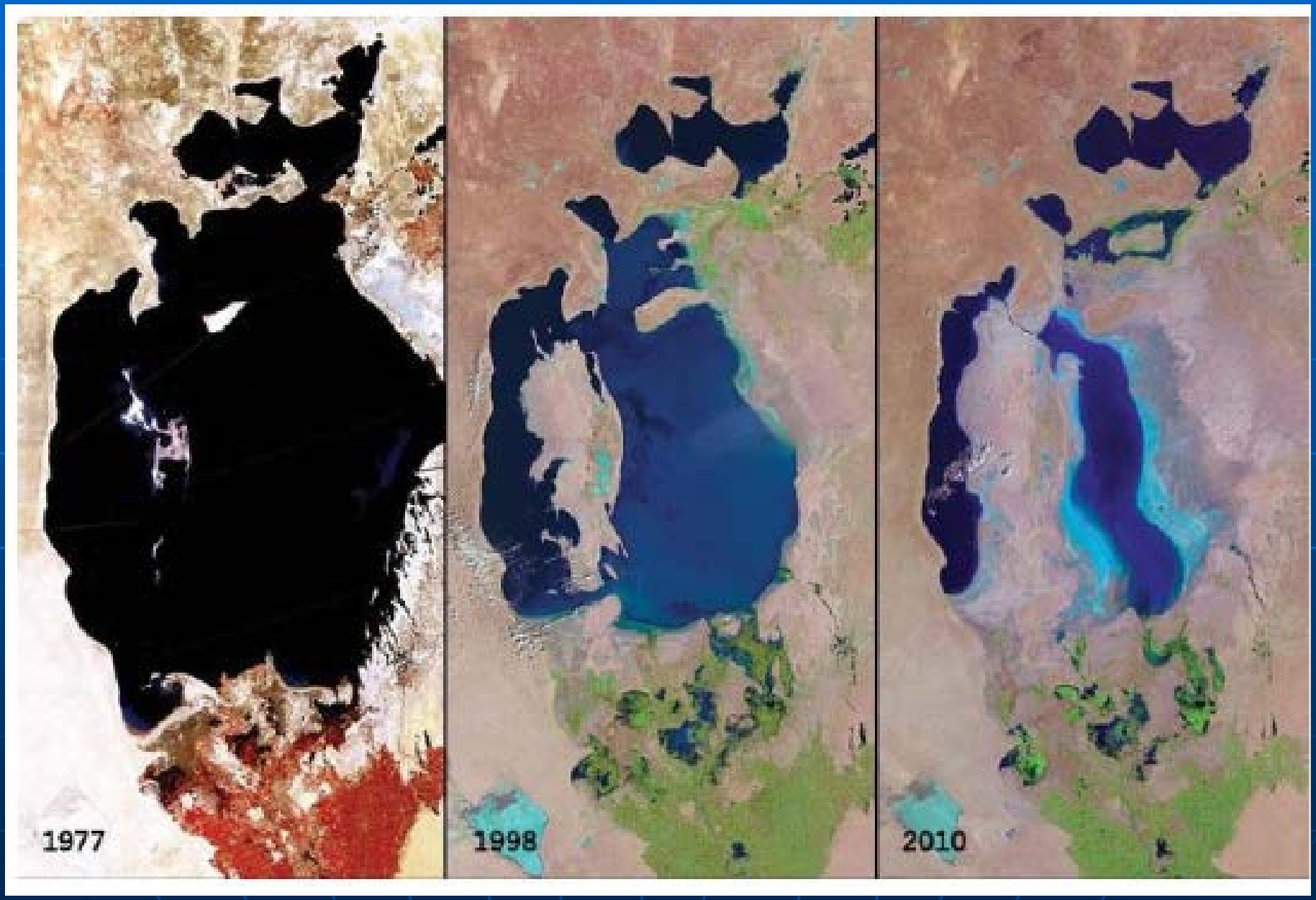
Spectral Reflectance –Temporal Effects



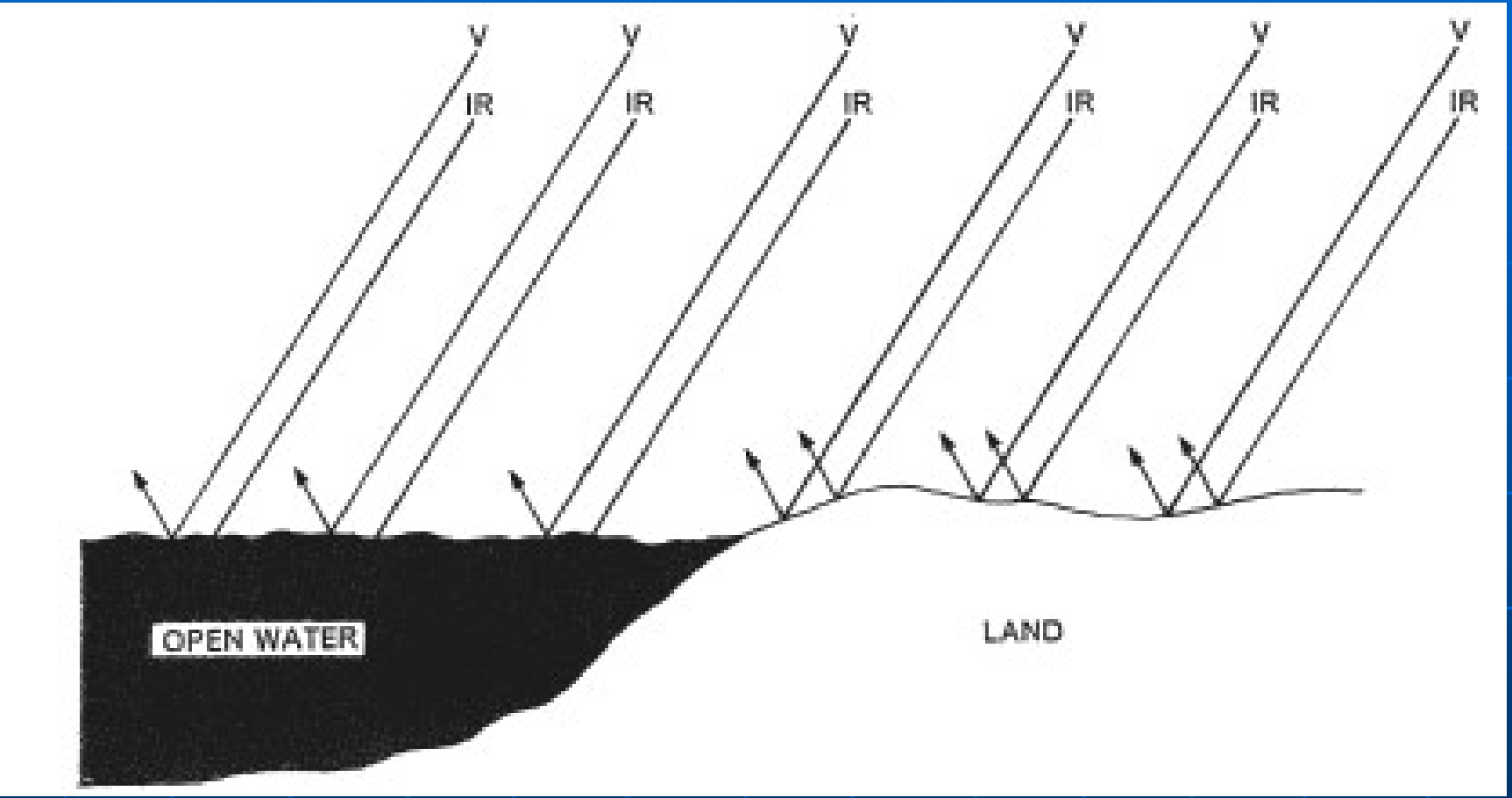
Sep 04 **Apr 05**

Dipalpur Canal provides water during summer only

Spectral Reflectance –Temporal Effects



Land-Water Boundary



Land–Water boundary: usually easiest in the near infrared region, where land, especially if vegetated, is bright, and open water dark

SPECTRAL REFLECTANCE OF SOIL

SOIL CHARACTERISTICS

- In RS data, soils may be distinguished by their varying spectral responses
- Texture and physical properties of soils results in unique spectral response
- The factors that influence spectral response includes:
  - Particle size (surface roughness)
  - Moisture content
  - Organic matter
  - Soil constituents (proportion of sand, silt, and clay),
  - Soil Color

SPECTRAL REFLECTANCE CHARACTERISTICS

- These factors are complex, variable, and interrelated. For example, the presence of moisture in soil will decrease its reflectance
- Soil moisture content is strongly related to the soil texture: coarse, sandy soils are usually well drained, resulting in low moisture content and relatively high reflectance
- Poorly drained fine textured soils will generally have lower reflectance.

SPECTRAL REFLECTANCE CHARACTERISTICS

- In the absence of water, the soil exhibits the reverse tendency:
  - coarse-textured soils will appear darker than fine-textured soils. Thus, the reflectance properties of a soil are consistent only within particular ranges of conditions.
- Organic matter reduces soil reflectance

SOIL CHARACTERISTICS

- Soils with organic matter content are often dark colored
- Moist soils have nearly uniform dark tones and low reflectance
- Dry soils give light tone and high reflectance
- Marshy areas can easily be identified due to high absorption of EMR of water and gives dark black tone
- Saline areas -high reflectance (white tone)

Moisture Content

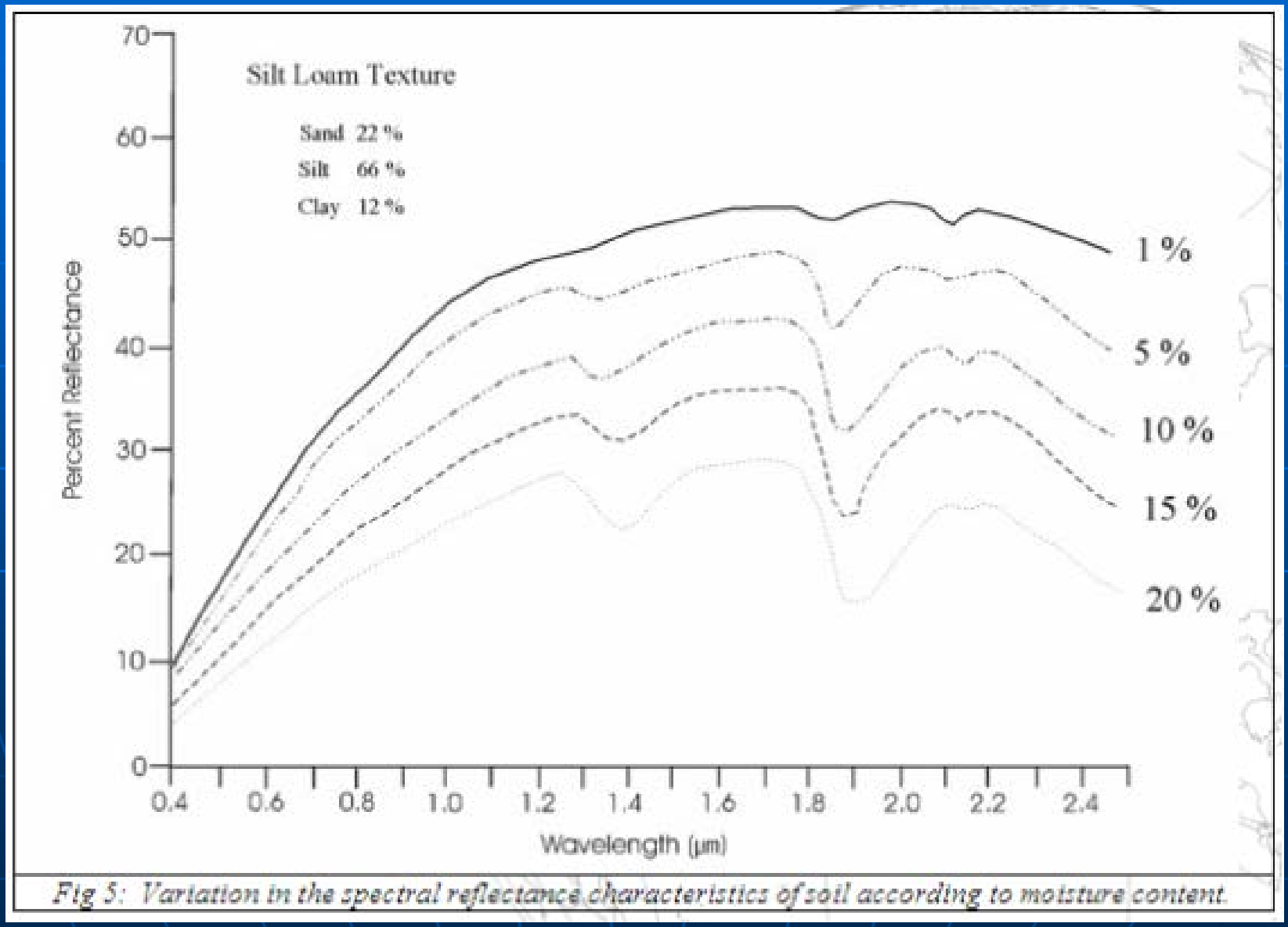


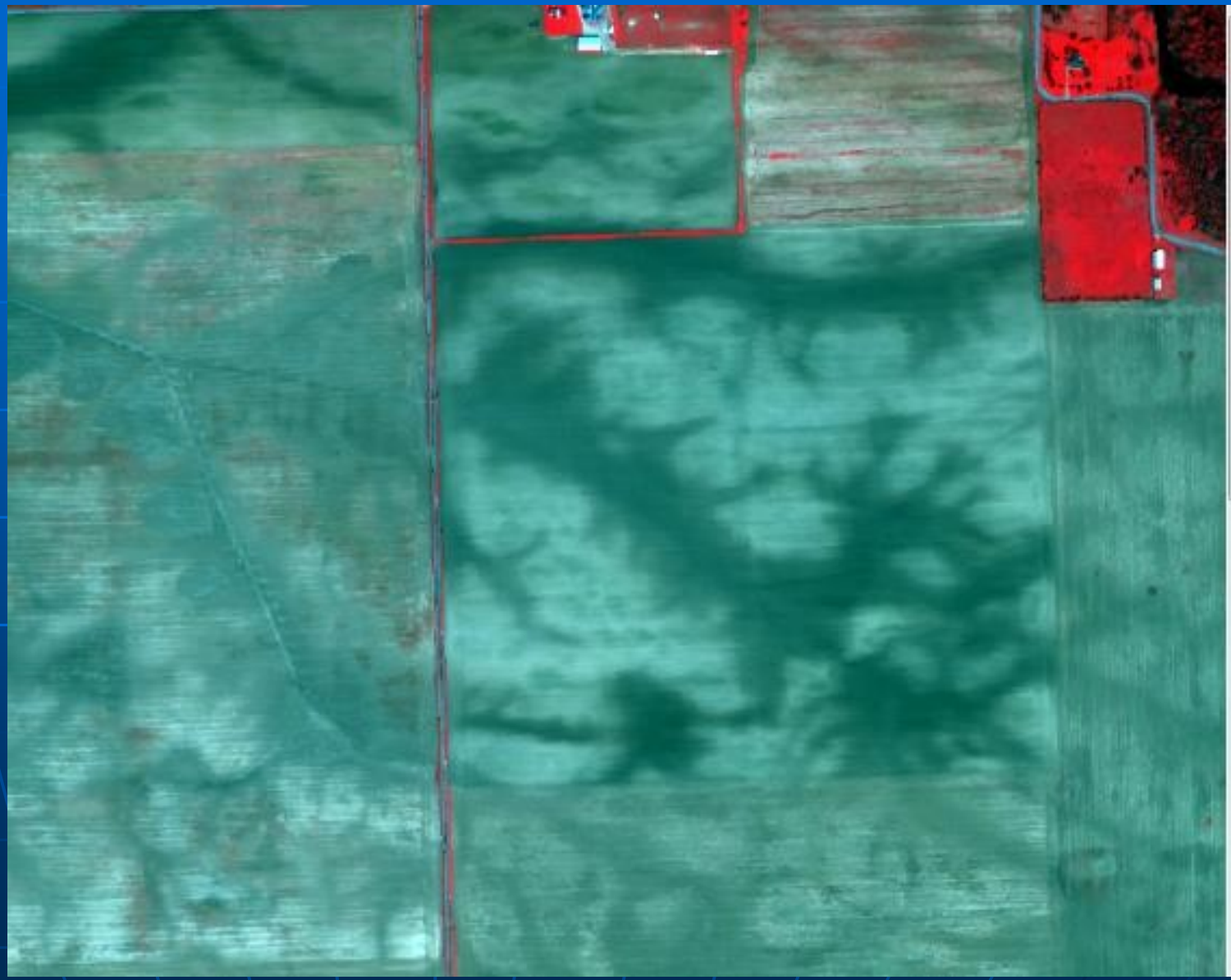
Fig 5: Variation in the spectral reflectance characteristics of soil according to moisture content.

Wet/Moist Soils



Wet/Moist Soils







Spectral Reflectance –Temporal Effects



Sep 04

Apr 05

Particle Size

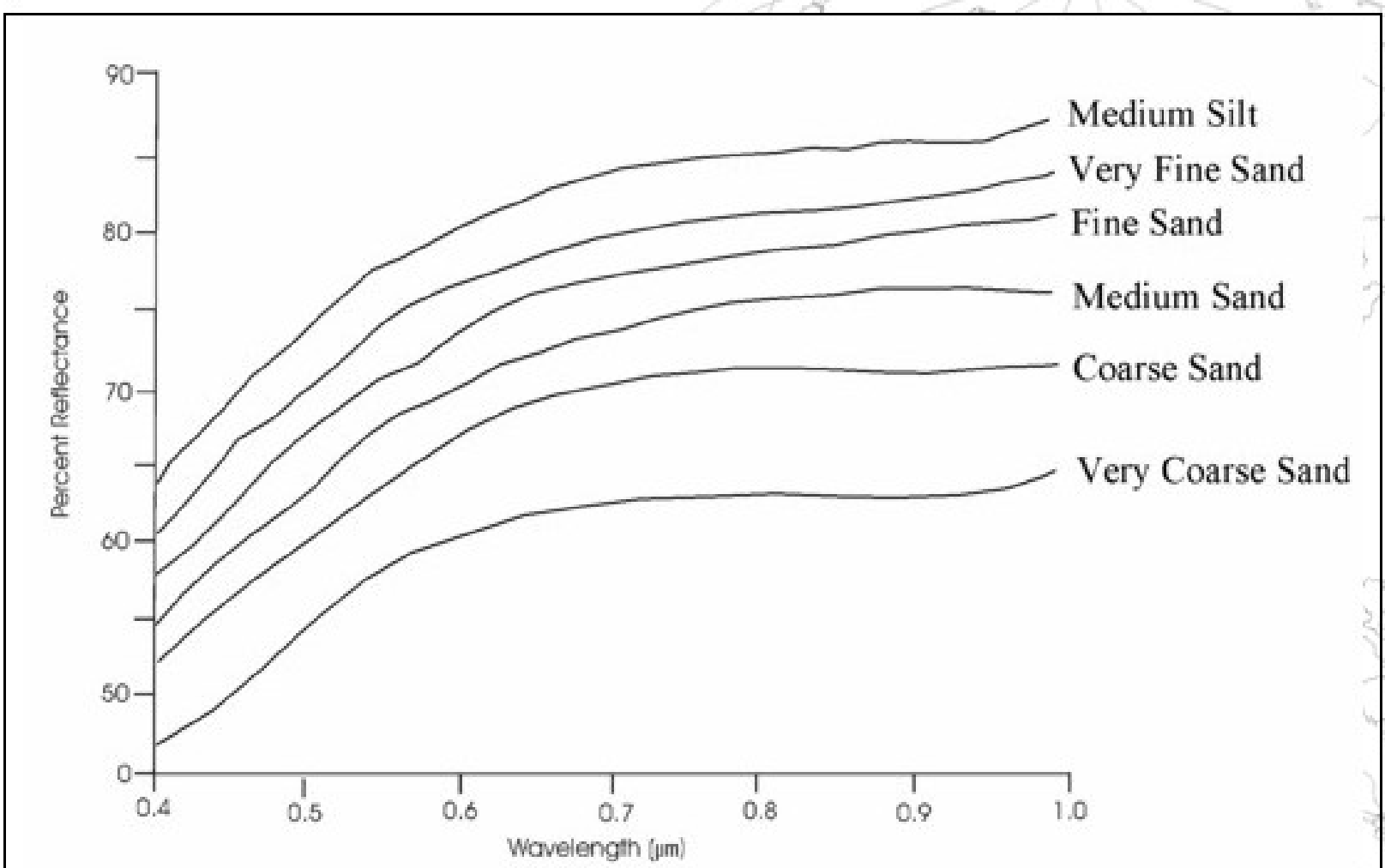
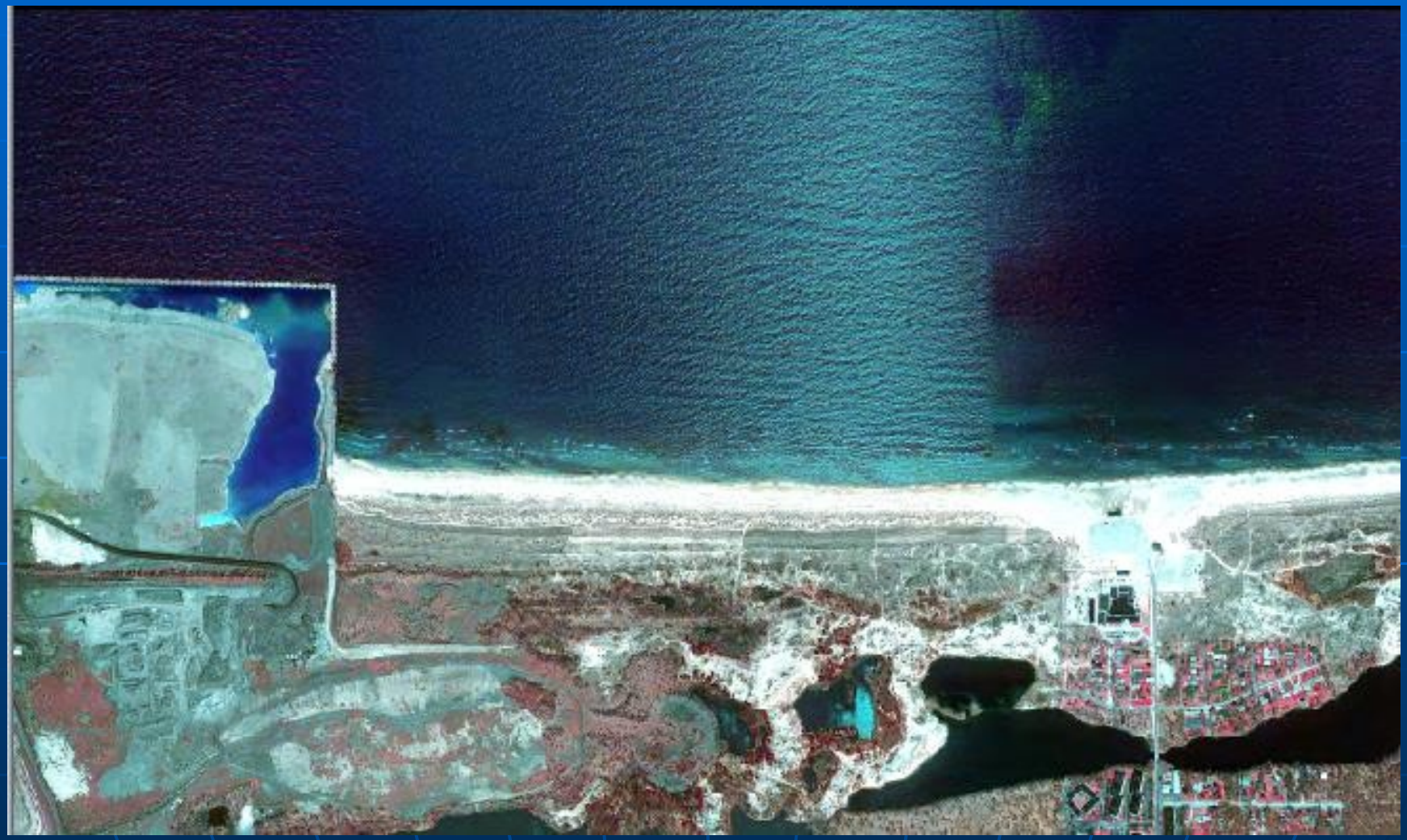


Fig 6: Variation in the spectral reflectance characteristics of soil according to particle size.

http://www.eps.umd.org/si/notes/m1r-1-8.htm

Surface reflectance –Sandy Areas



Soil Texture

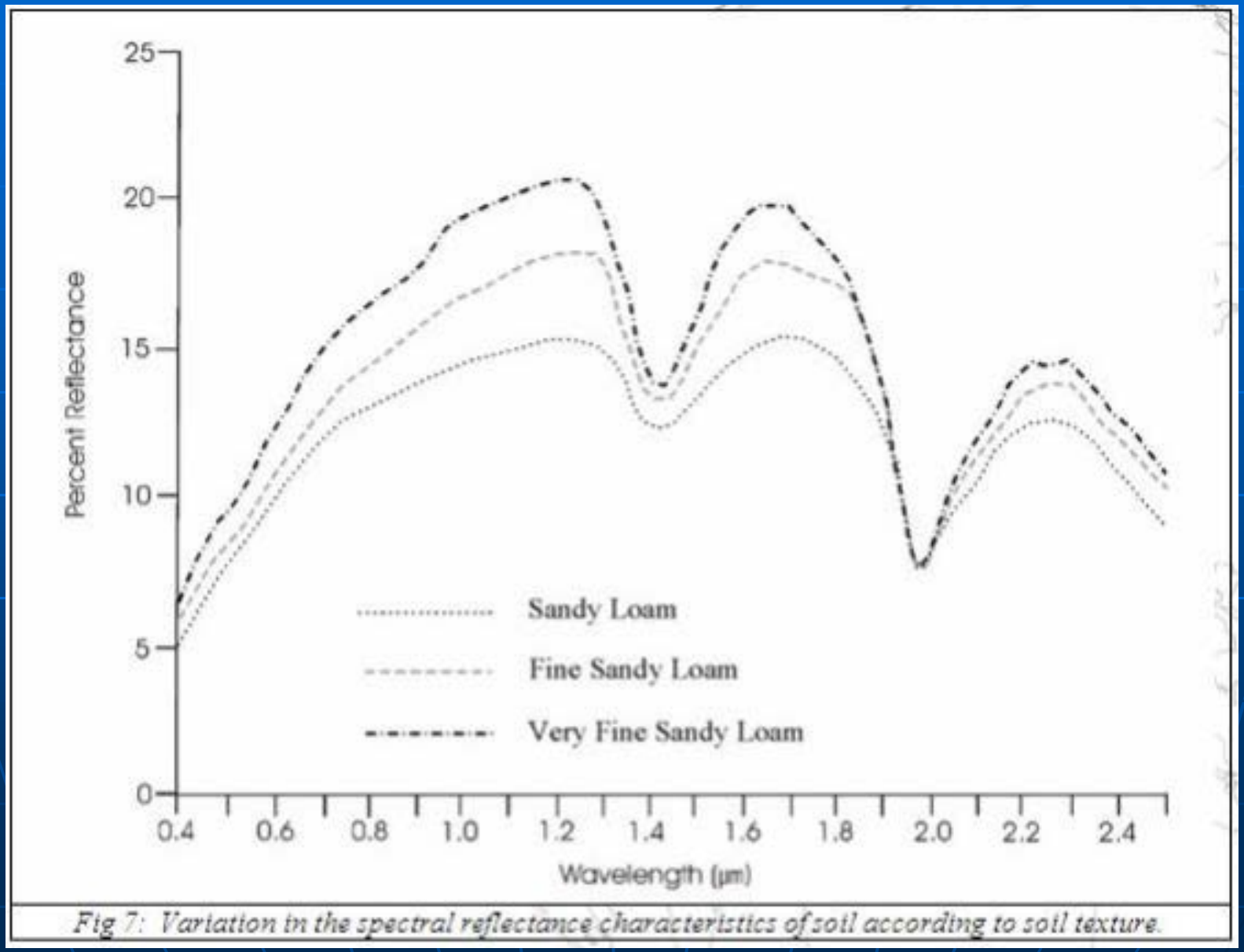


Fig 7: Variation in the spectral reflectance characteristics of soil according to soil texture.

SPECTRAL REFLECTANCE OF VEGETATION

INTERACTION OF EMR WITH VEGETATION

- Vegetation show variations that relates to geology depending upon the climate and soil
- Plants use solar energy to convert water and carbon dioxide to carbohydrate and oxygen through the process of photosynthesis with strong influence on their interaction with EMR

INTERACTION OF EMR WITH VEGETATION

- The catalyst for photosynthesis is pigment chlorophyll
- The function of chlorophyll is to absorb solar radiations and there by fuel photosynthesis
- This is achieved by absorption near **0.45 and 0.68 µm** in the blue and red part of the visible spectrum, *which is why healthy leaves appear* ***green***

INTERACTION OF EMR WITH VEGETATION

- Water in the cells does absorb some energy
- Chlorophyll, water content and surface morphology of different plants have a marked effect on their spectral properties in the visible to NIR spectrum

INTERACTION OF EMR WITH VEGETATION

- Healthy vegetation look very similar in the visible part of the spectrum but can be distinguished in NIR
- Plants leaves, spaces, and branches with different leaf shapes and sizes can broaden the range of response, eases discrimination between species and healthy and weak members of the same species

INTERACTION OF EMR WITH VEGETATION

- Vegetation have life cycles of various durations thus, their spectral properties are not fixed
- As a deciduous leaf matures before falling, its chlorophyll contents decay away removing the strong red absorption consequently, it changes its color from green to yellow to red
- As cell shrinks and dry, they become less efficient at reflecting NIR

INTERACTION OF EMR WITH VEGETATION

- As the leaves fall, less of the plant intercepts solar radiation and the reflectance becomes dominated by the soil and leaf litter beneath it
- When new leaves appear , NIR reflectance is well developed coupled with high yellow reflectance

INTERACTION OF EMR WITH VEGETATION

- Coniferous trees do not shed their leaves during one season, and always display high chlorophyll abundance
- Since their leaves are small less solar radiation is intercepted so the reflectance of the whole plant is lower than the deciduous trees with full leaf





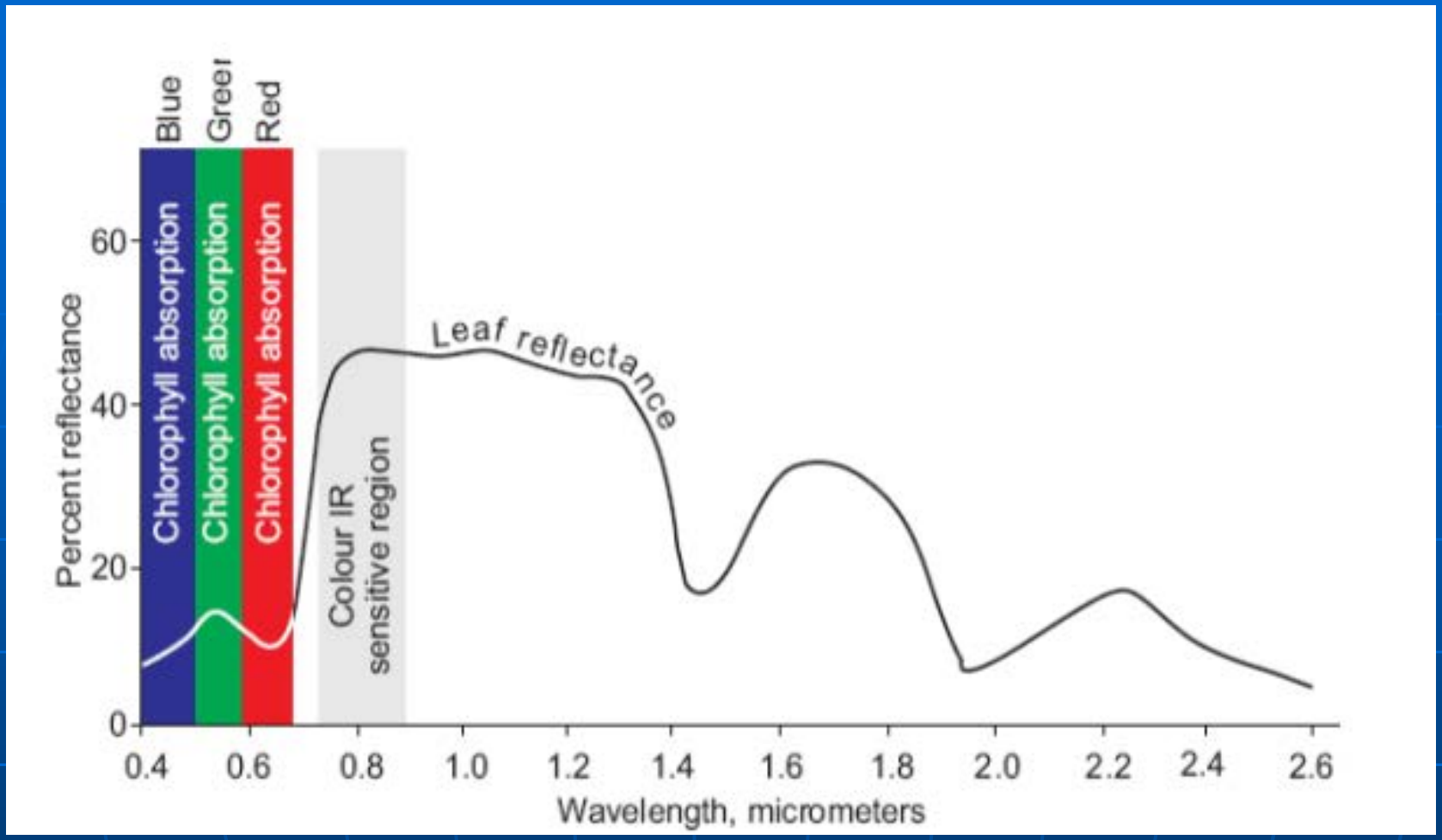


GREEN VEGETATION

- Relative reflectance is higher in the visible green than in blue and red ,thus giving the vegetation visible green color
- Reflectance is much higher in reflective IR region than visible
- This high IR reflectance is something unique to green vegetation (live) as compared to green netting thrown over a tank-a camouflage detection property

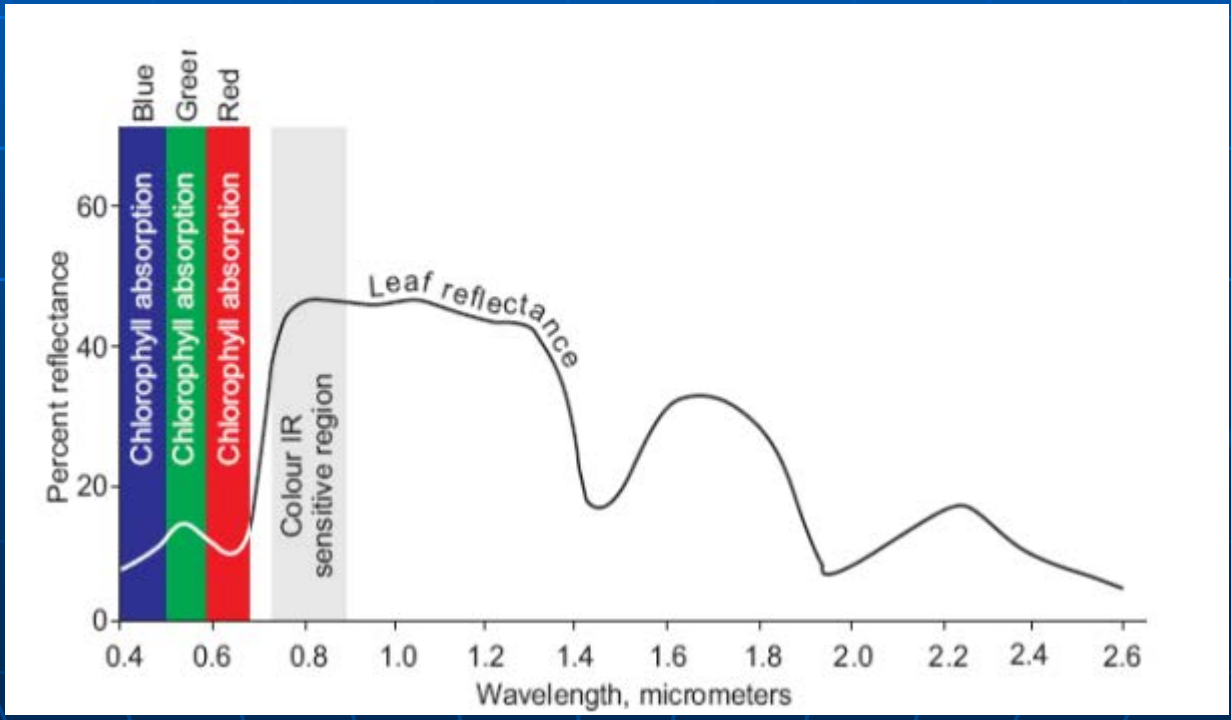
SPECTRAL REFLECTANCE CHARACTERISTICS

- Dead vegetation reflects greater amount of energy than healthy green vegetation throughout the visible spectrum (0.4-0.7 µm)
- Conversely, it reflects less in the reflective nir region (0.7-1.1 µm)

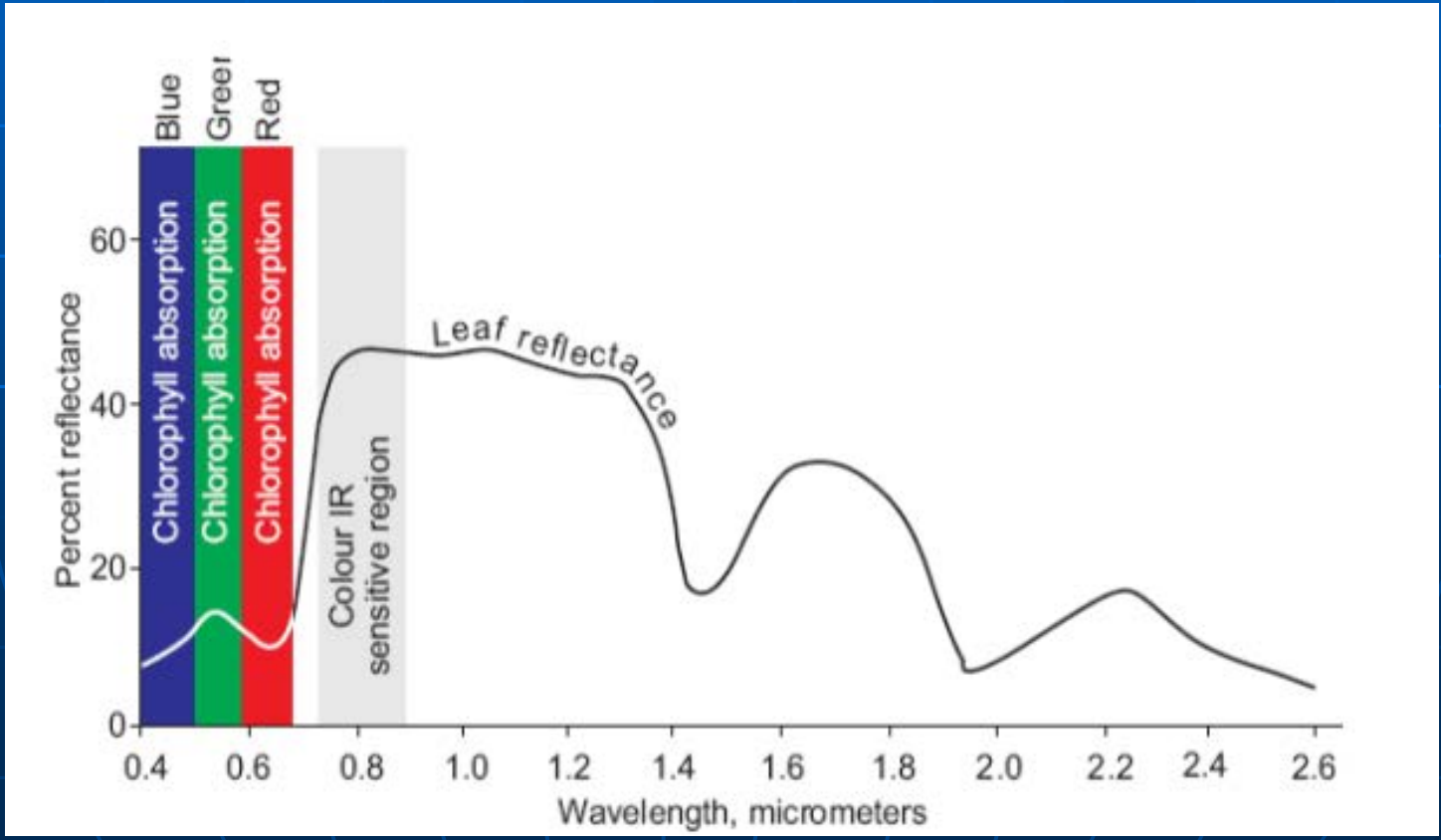


Strong absorption in visible and high reflectance in NIR
Chlorophyll, strongly absorbs energy in the wavelength bands at about 0.45 and 0.67 μm ,often called the "Chlorophyll Absorption Bands"

- If a plant is subject to some form of stress that interrupts its normal growth and productivity, it may decrease or cease chlorophyll production.
- This result is less chlorophyll absorption in the blue and red bands. Often, the red reflectance increases to the point that we see the plant turn yellow (combination of green and red)



- From the visible to the near-IR portion of the spectrum at about 0.7 μm , the reflectance of healthy vegetation increases dramatically .
- In the range from about 0.7 to 1.3 μm , a plant leaf typically reflects 40 to 50 percent of the energy incident upon it.



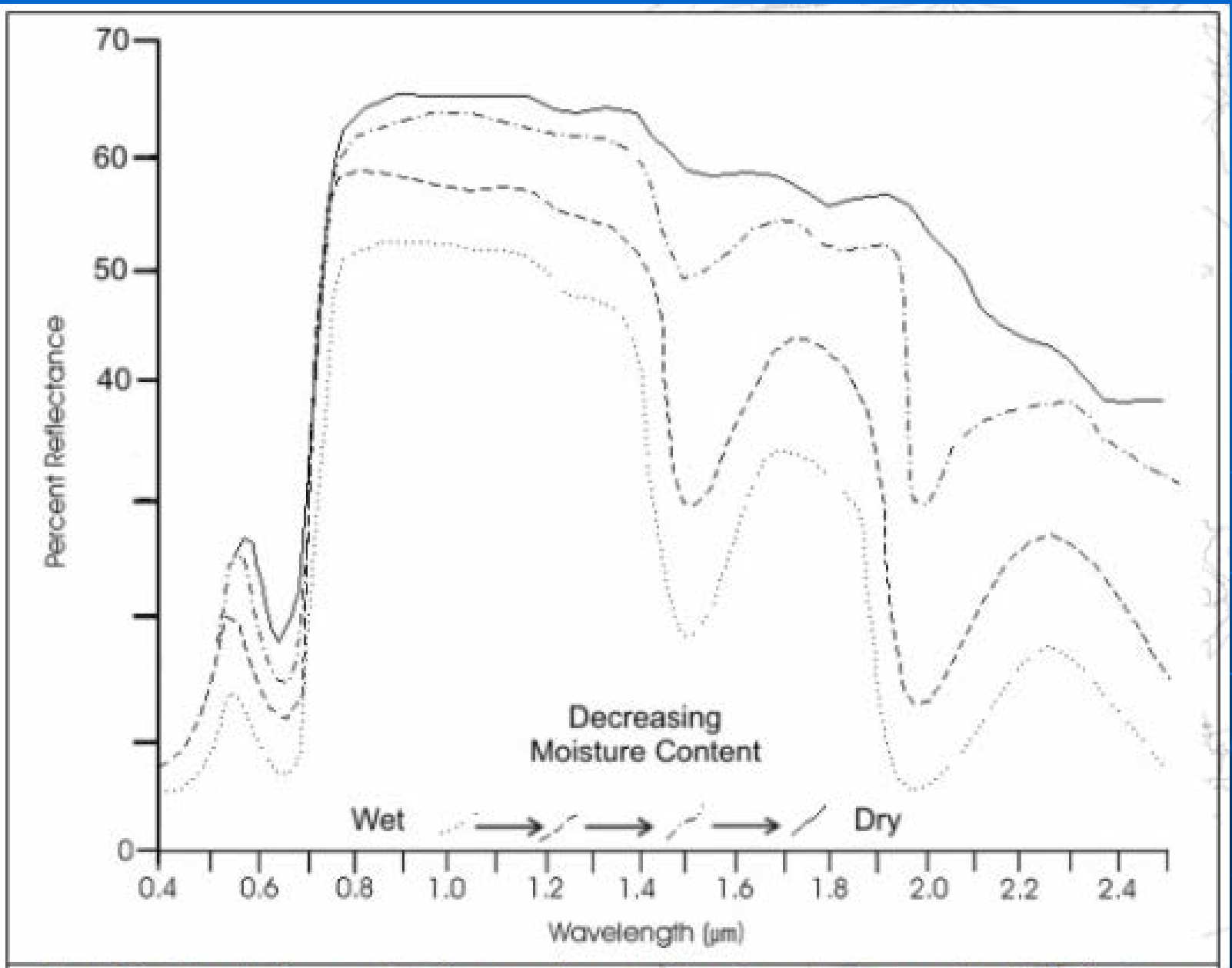
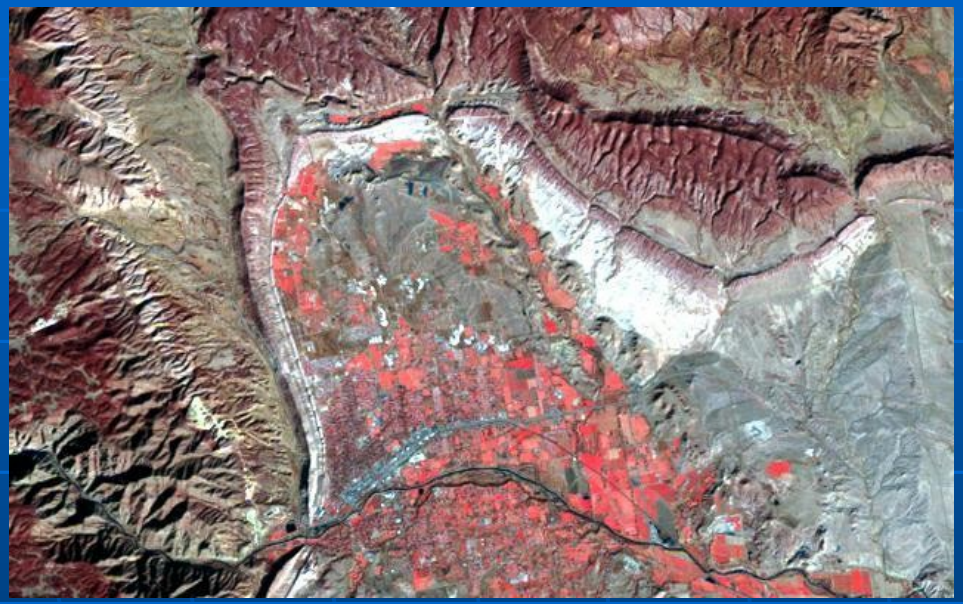
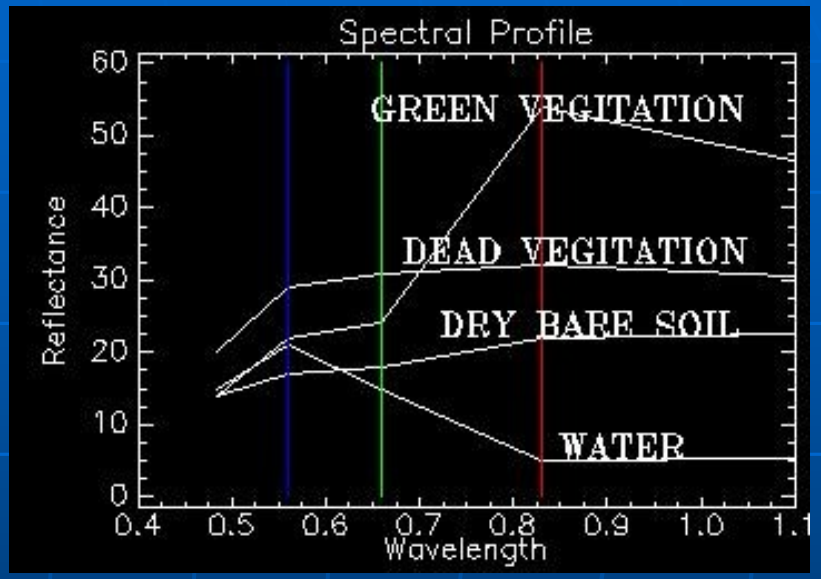


Fig 4: Variation in the spectral reflectance characteristics of vegetation according to leaf moisture content.

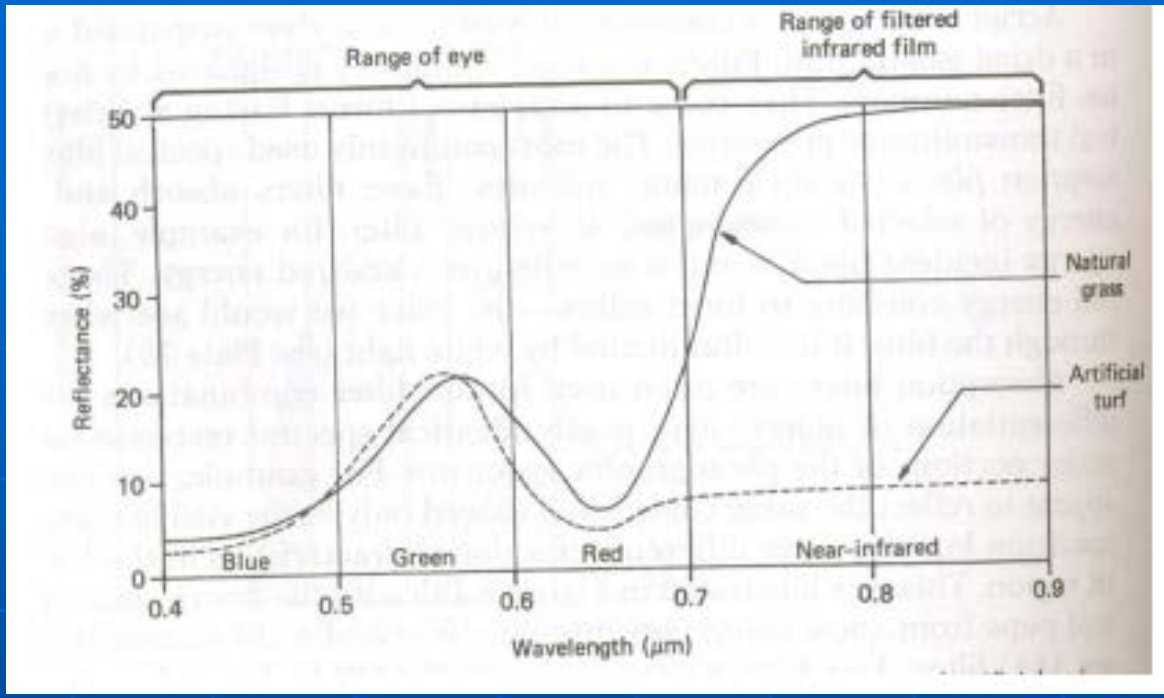
IMAGE AND SPECTRAL PROFILE







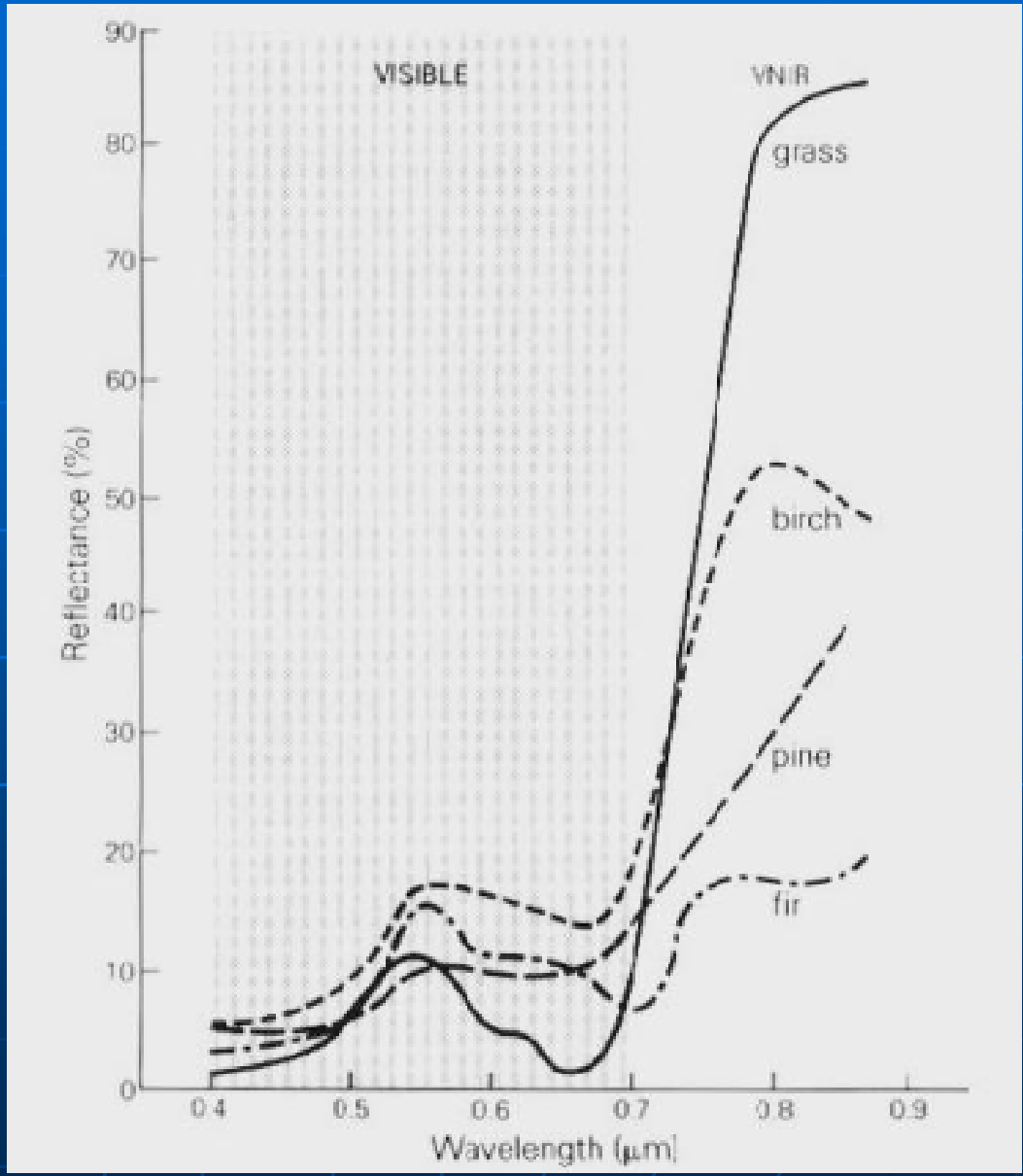




Spectral reflectance curves for natural grass and turf artificially made of green color to visually resemble green grass, reflectance in blue, red and green is similar for both Surfaces

In NIR natural grass reflects very high but turf does not

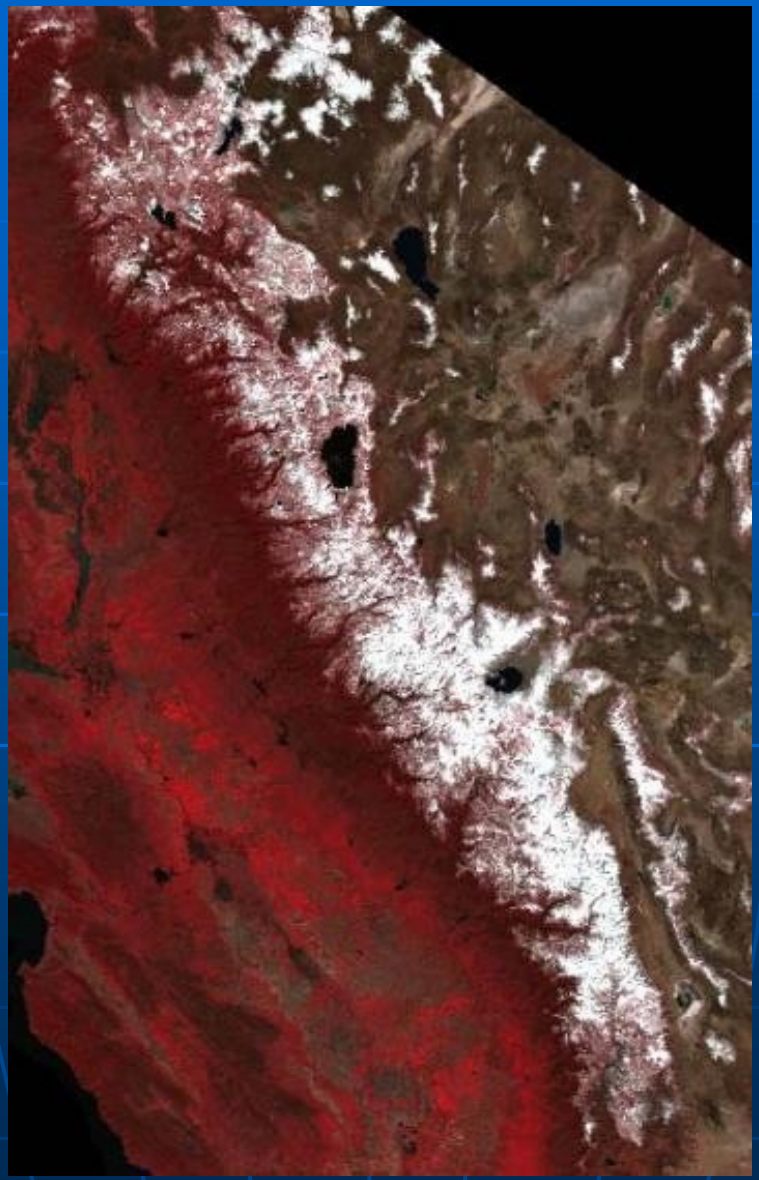
All of the different attributes Of chlorophyll content, leaf shape, area and number together with plant structure, contributes
to spectral reflectance of plant species
All four plant have similar response in visible but are clearly distinguished in by their NIR reflectance



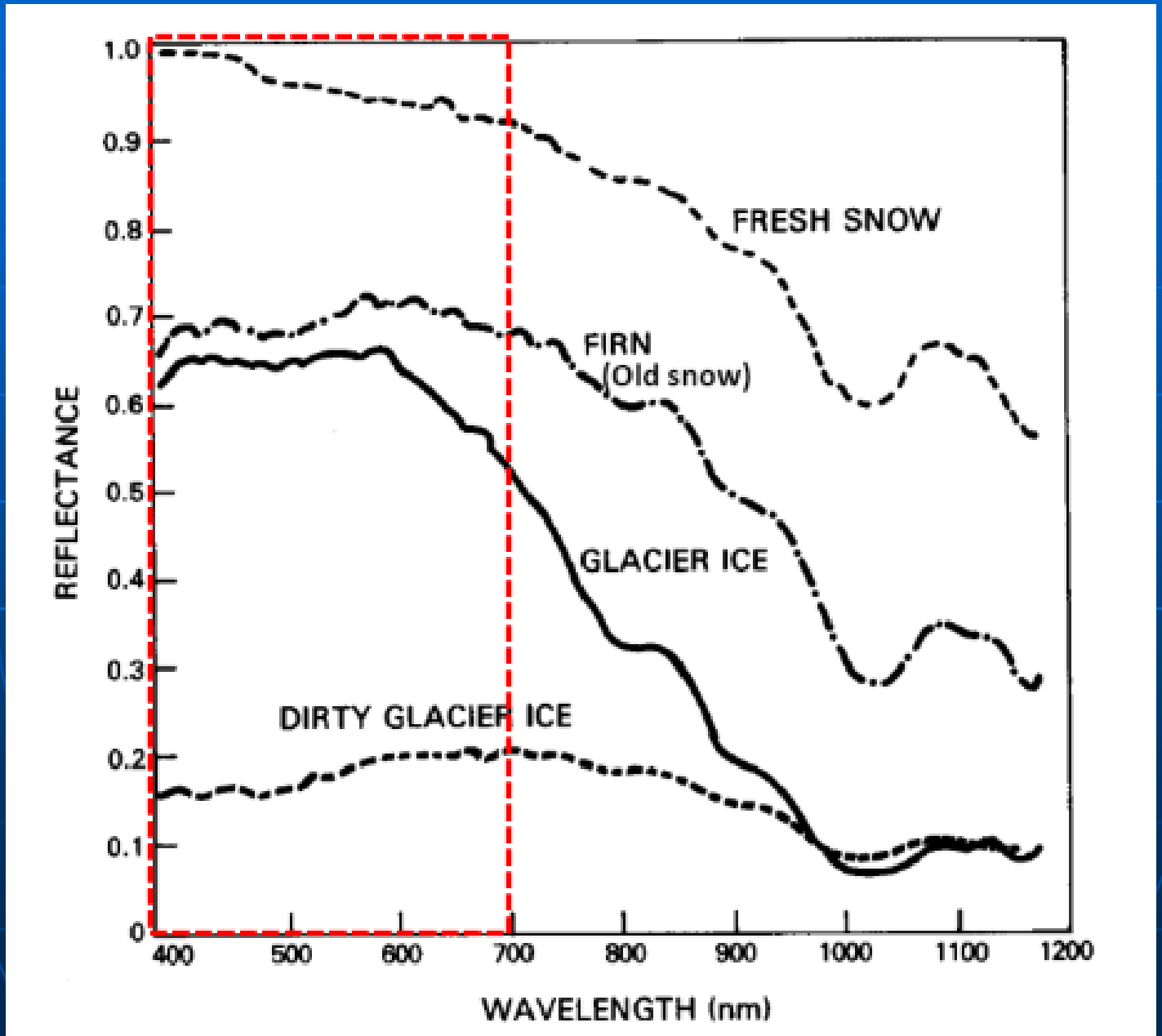
Spectral Reflectance- Snow

- The reflectance and transmittance of the snowpack depend on the
  - Wavelength,
  - Grain size distribution of the snow,
  - Depth and density of the snowpack,
  - Size and amount of the impurities

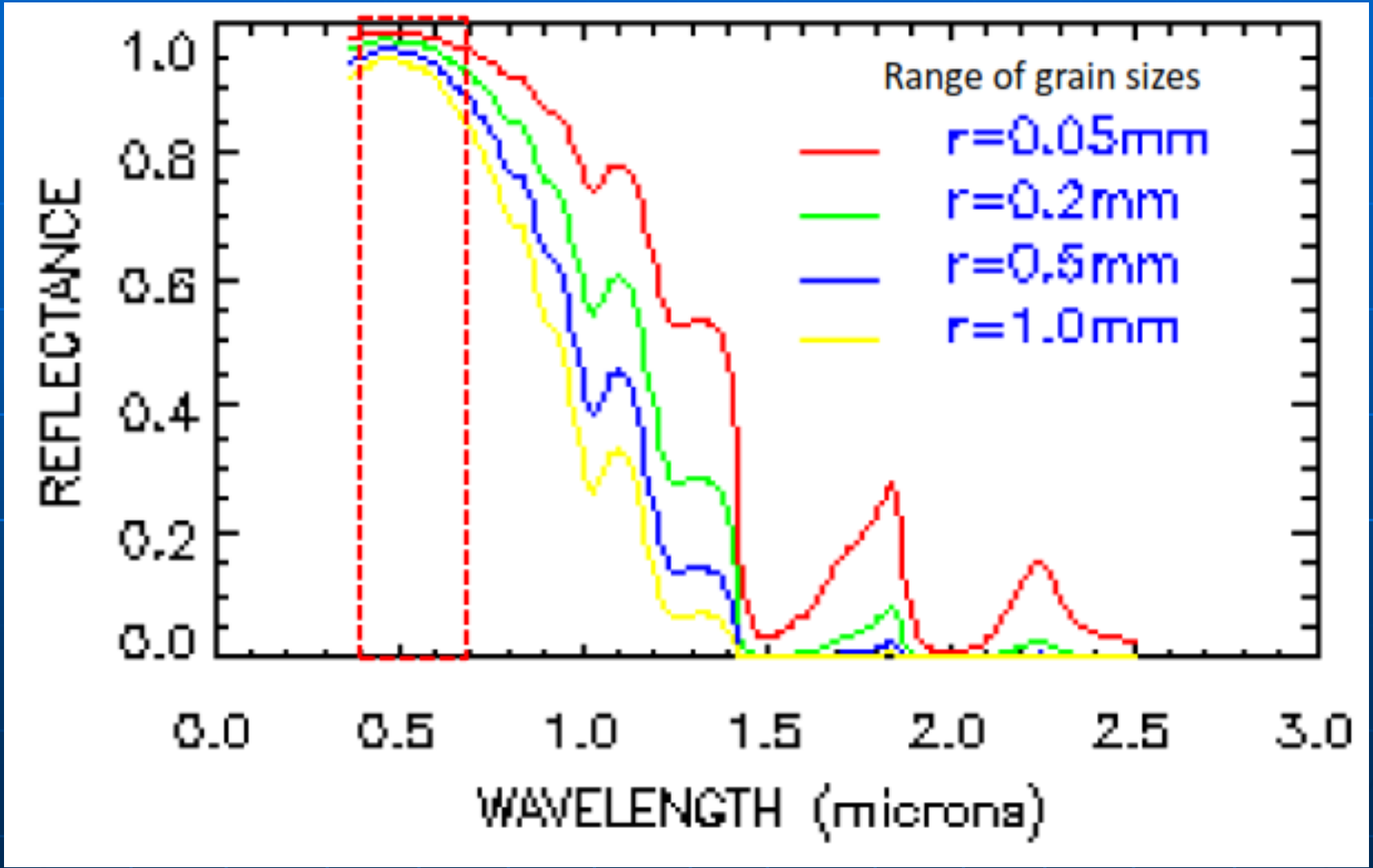
Spectral Reflectance- Snow



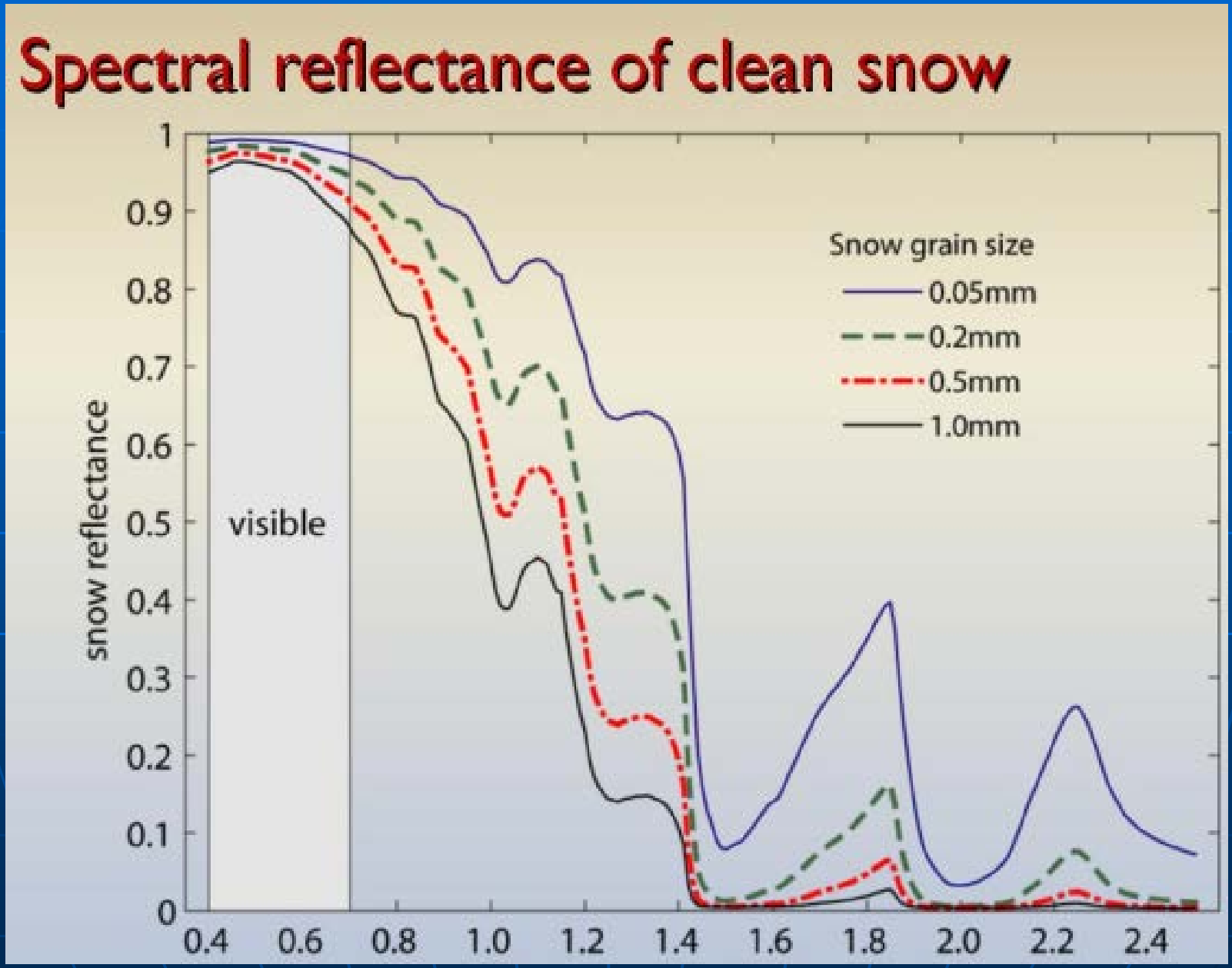
Spectral Reflectance- Snow



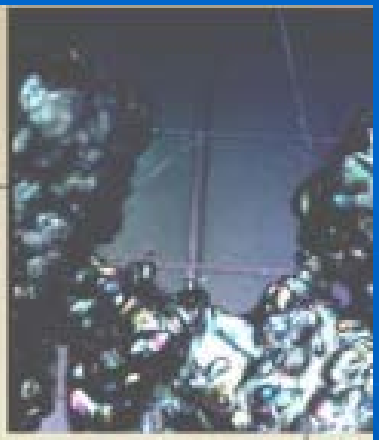
Spectral Reflectance- Snow

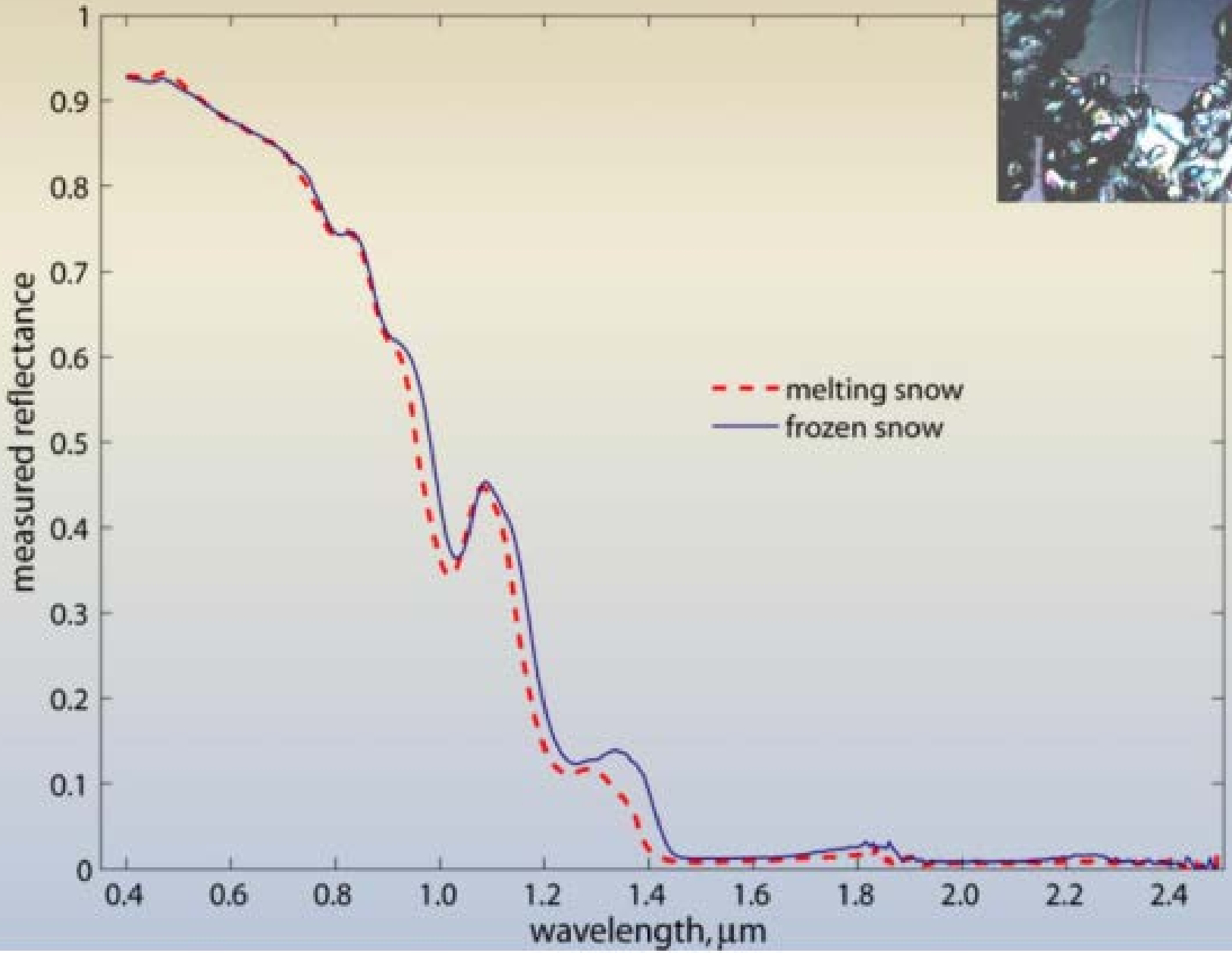


Spectral reflectance of clean snow



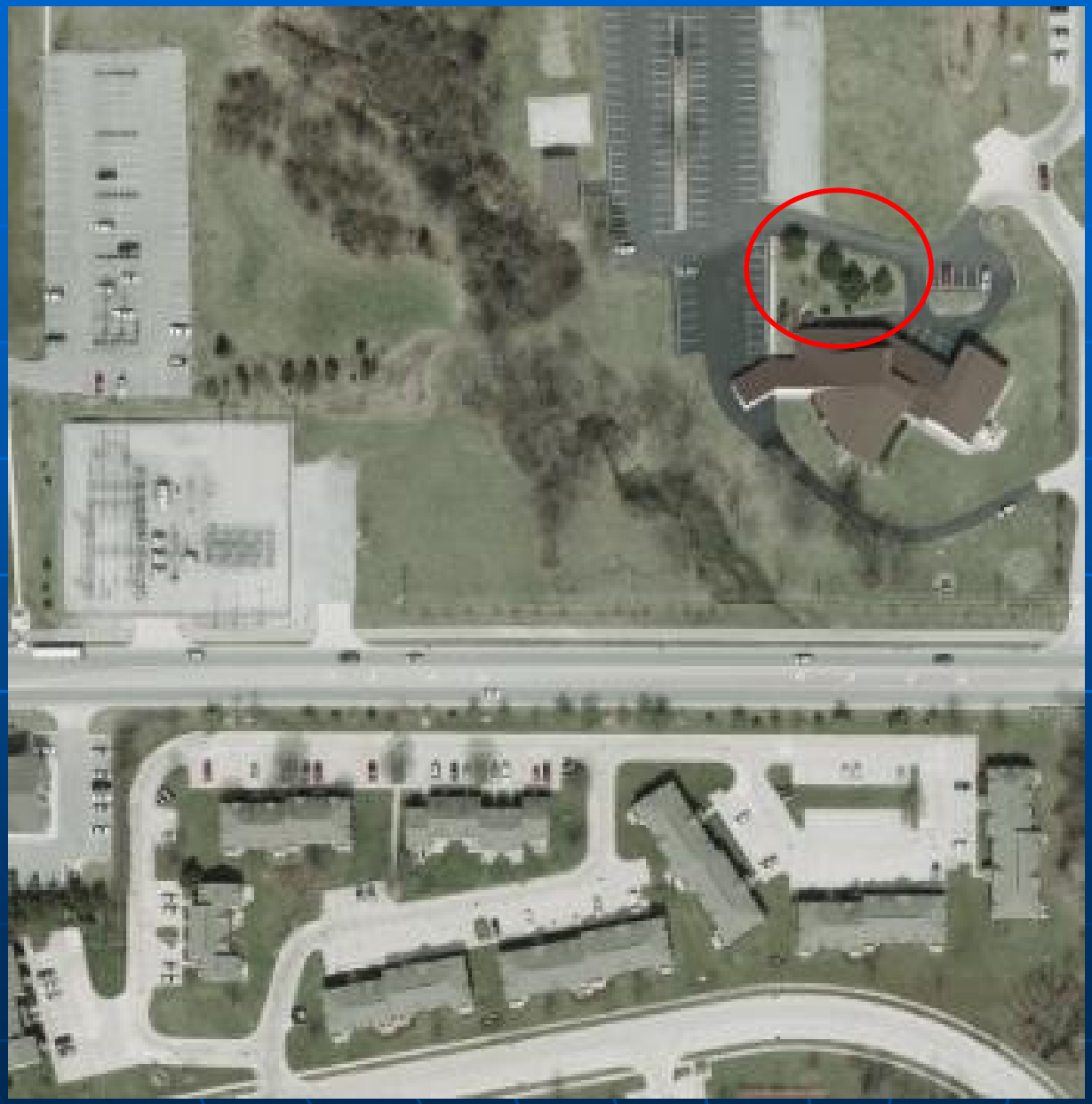
Wet snow




measured reflectance
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
wavelength, μm
melting snow
frozen snow

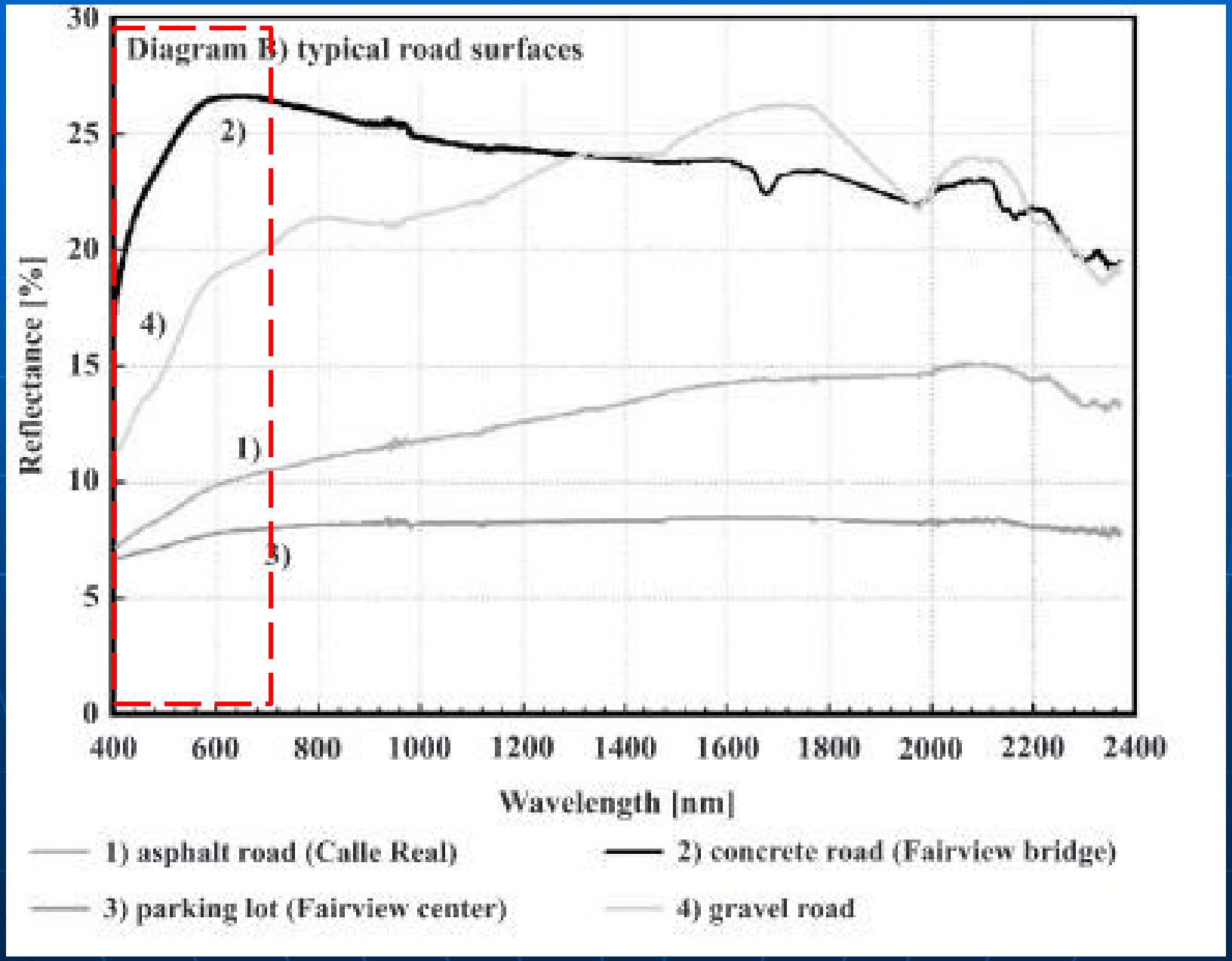
Impervious Surfaces

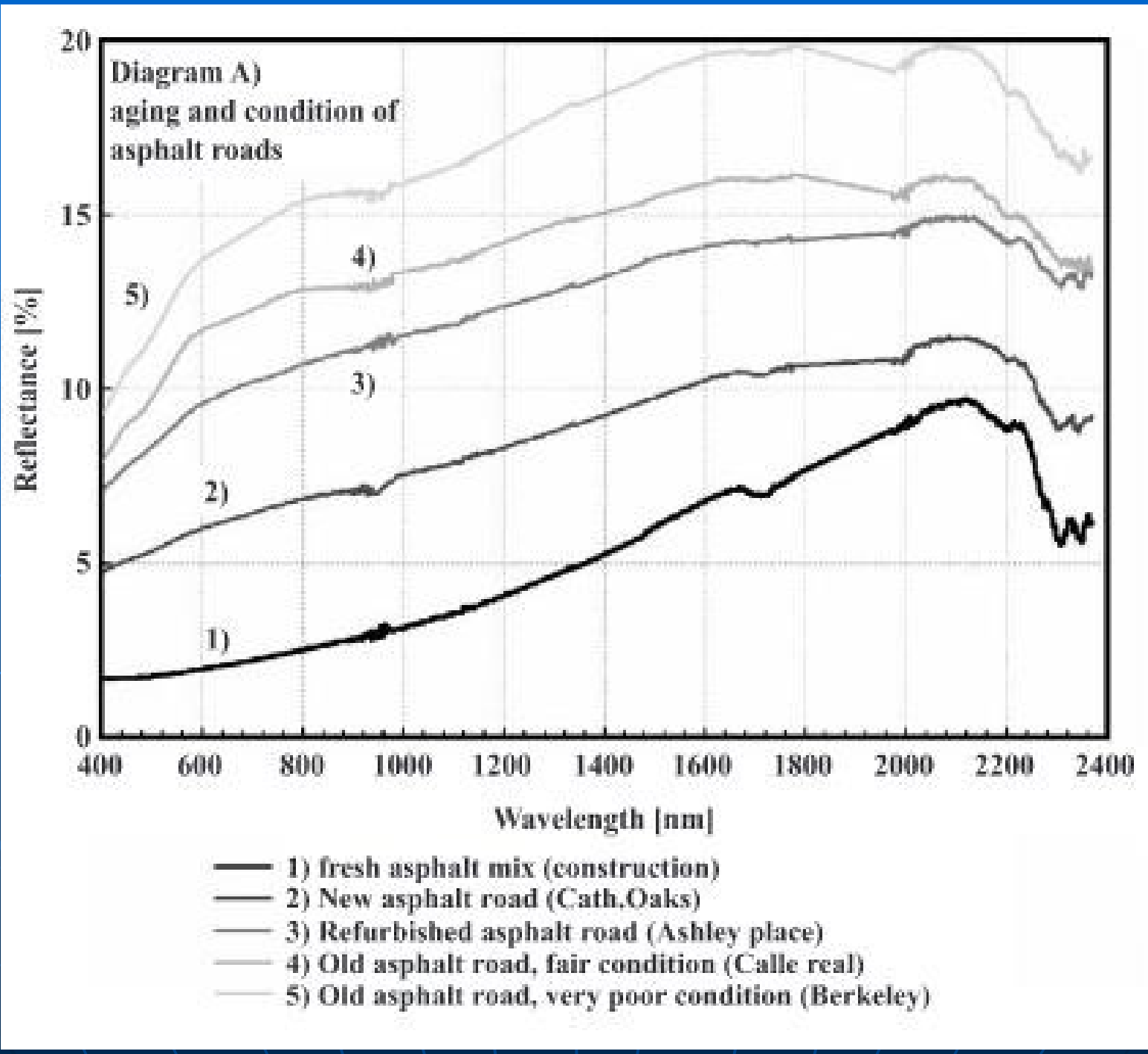


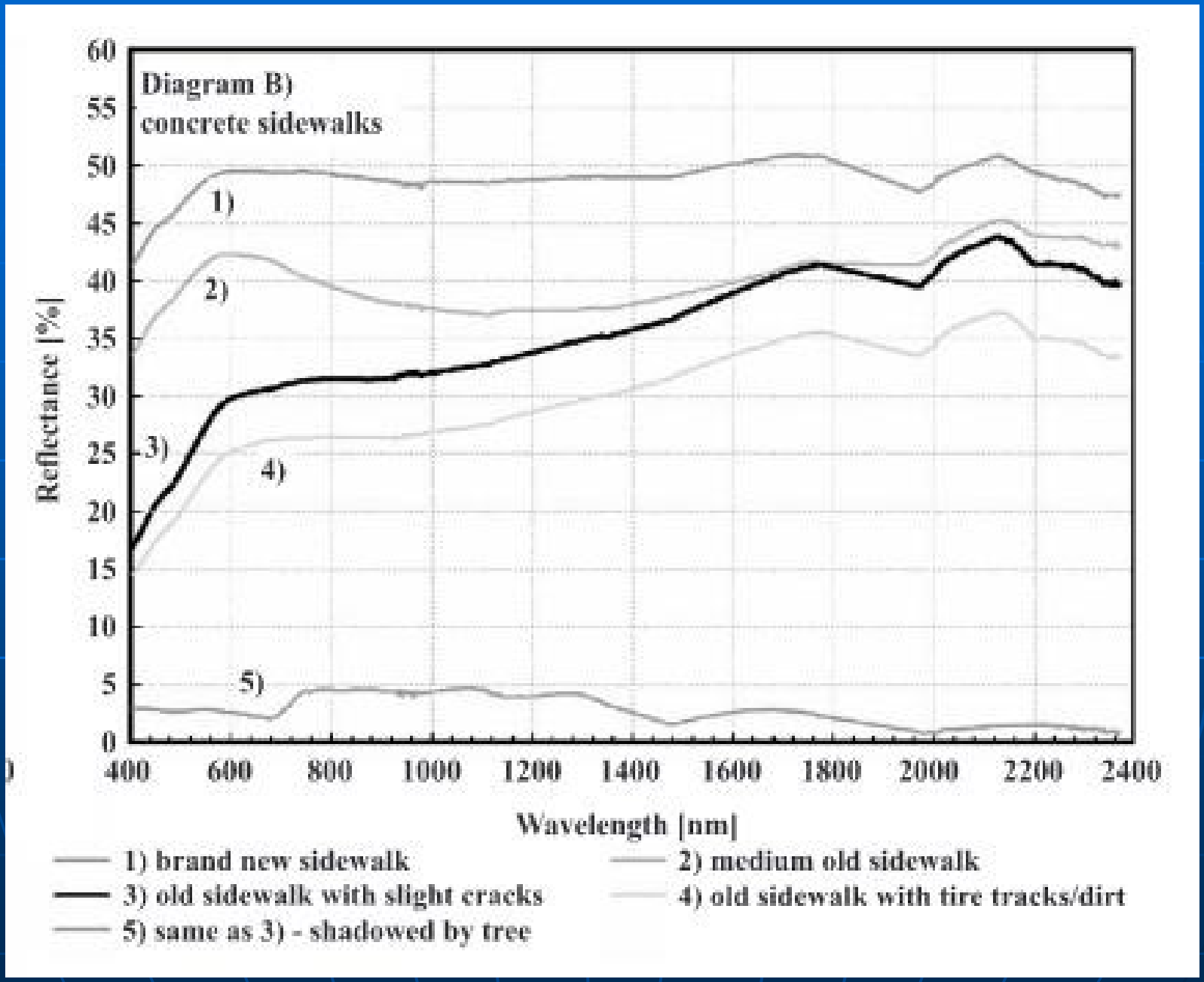


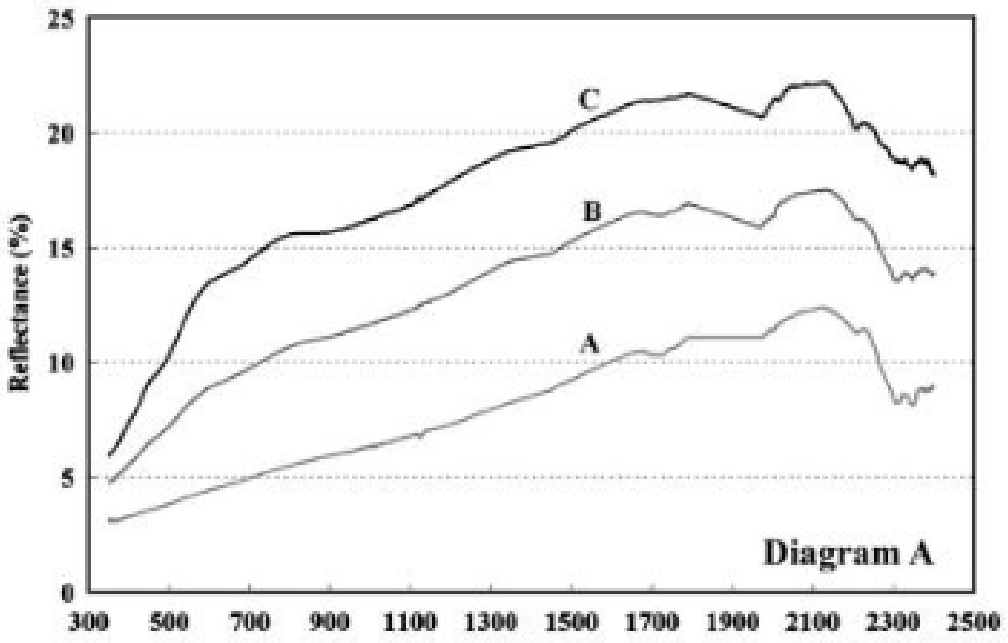


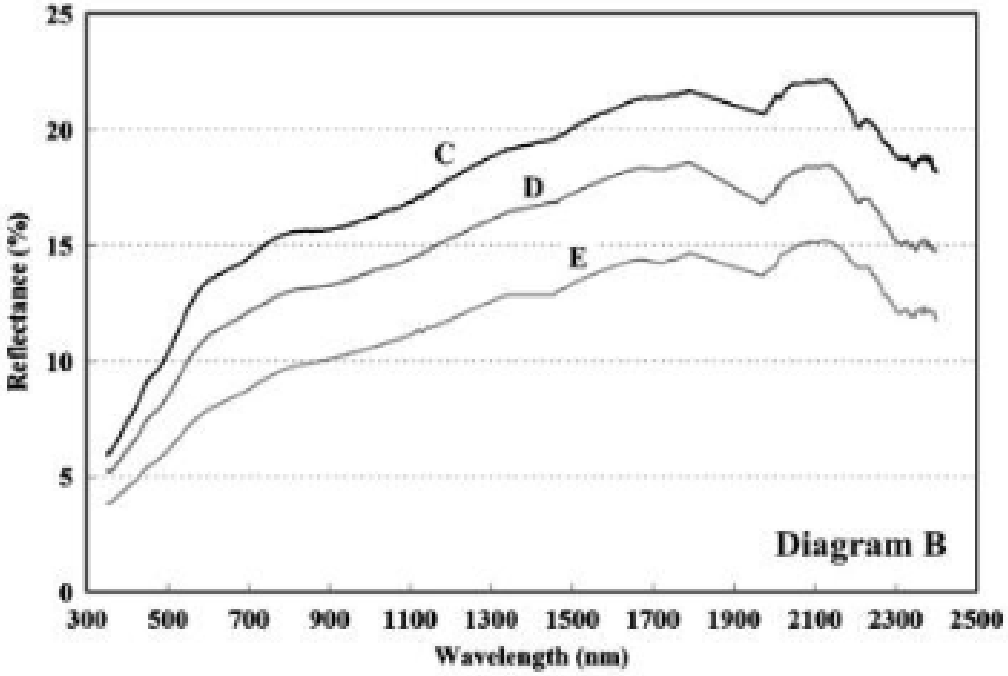
Spectral Reflectance- Road and Parking Surfaces

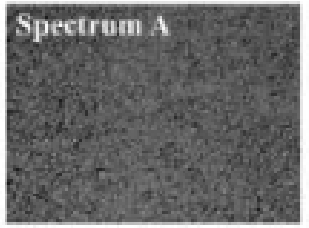



Diagram A)
aging and condition of
asphalt roads
Reflectance [%]
20
15
10
5
0
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
Wavelength [nm]
1)
2)
3)
4)
5)
1) fresh asphalt mix (construction)
2) New asphalt road (Cath.Oaks)
3) Refurbished asphalt road (Ashley place)
4) Old asphalt road, fair condition (Calle real)
5) Old asphalt road, very poor condition (Berkeley)

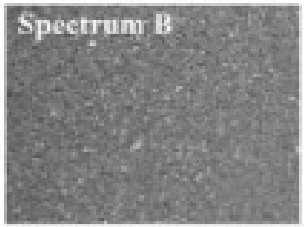

Diagram B)
concrete sidewalks
Reflectance [%]
60
55
50
45
40
35
30
25
20
15
10
5
0
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
Wavelength [nm]
1)
2)
3)
4)
5)
1) brand new sidewalk
2) medium old sidewalk
3) old sidewalk with slight cracks
4) old sidewalk with tire tracks/dirt
5) same as 3) - shadowed by tree



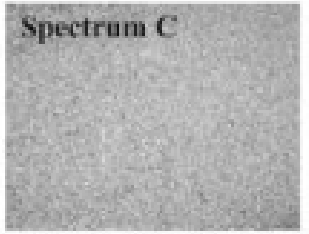




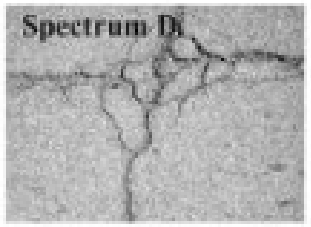
Age:<1yr/PCI:99/SI:100



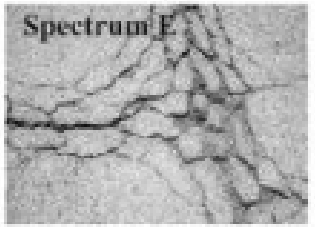
Age:3yrs/PCI:86/SI:100



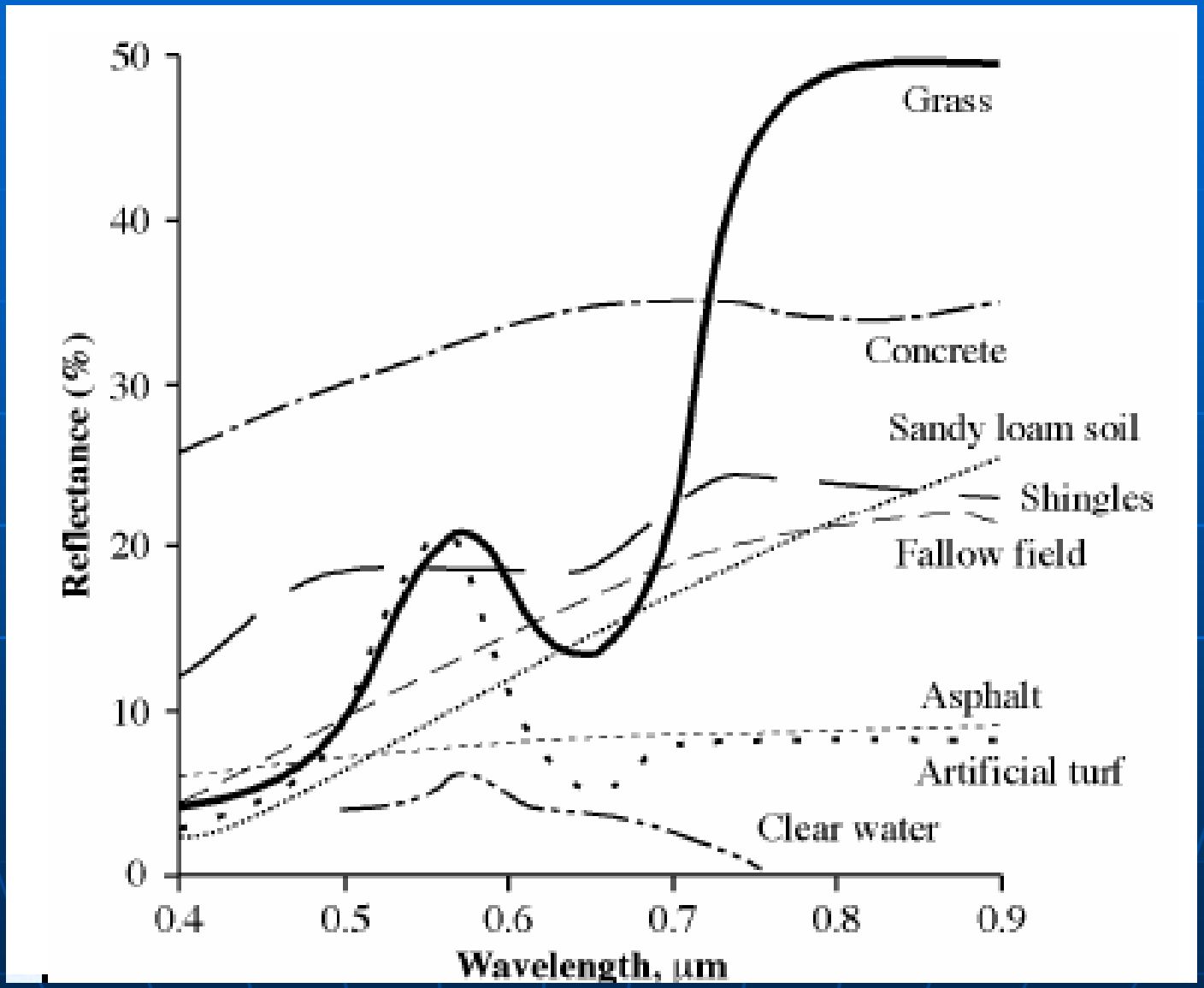
Age:>10yrs/PCI:32/SI:63



Low severity crack



Severe alligator crack


50
40
30
20
10
0
Reflectance (%)
0.4
0.5
0.6
0.7
0.8
0.9
Wavelength, μm
Grass
Concrete
Sandy loam soil
Shingles
Fallow field
Asphalt
Artificial turf
Clear water

Reference and Reading Assignment

- Paul M Mather, Chapter 1, section 1.3.2 Spectral reflectance of Earth surface materials

  Remote Sensing of Environment by Jensen, Chapter Remote sensing of the Vegetation